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Vector Algebra

JEE Main Mathematics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 194 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Let $\vec{a} = 4\hat{i} - \hat{j} + 3\hat{k}$, $\vec{b} = 10\hat{i} + 2\hat{j} - \hat{k}$ and a vector $\vec{c}$ be such that $2(\vec{a} \times \vec{b}) + 3(\vec{b} \times \vec{c}) = \vec{0}$.

If $\vec{a} \cdot \vec{c} = 15$, then $\vec{c} \cdot (\hat{i} + \hat{j} - 3\hat{k})$ is equal to :

(A) -6 (B) -5 (C) -4 (D) -3

Open solution

Q2. JEE Main 2026 (Online) 6th April Evening Shift

Let $\vec{a} = 2\hat{i} + 3\hat{j} + 3\hat{k}$ and $\vec{b} = 6\hat{i} + 3\hat{j} + 3\hat{k}$. Then the square of the area of the triangle with adjacent sides determined by the vectors $(2\vec{a} + 3\vec{b})$ and $(\vec{a} - \vec{b})$ is :

(A) 450 (B) 900 (C) 1800 (D) 2400

Open solution

Q3. JEE Main 2026 (Online) 5th April Evening Shift

Let O be the origin, $\overrightarrow{OP} = \vec{a}$ and $\overrightarrow{OQ} = \vec{b}$. If R is the point on $\overrightarrow{OP}$ such that $\overrightarrow{OP} = 5\overrightarrow{OR}$, and M is the point such that $\overrightarrow{OQ} = 5\overrightarrow{RM}$, then $\overrightarrow{PM}$ is equal to :

(A) $\frac{1}{5}(\vec{a} - 4\vec{b})$ (B) $\frac{1}{5}(\vec{b} - 4\vec{a})$ (C) $\frac{1}{5}(-\vec{a} + 4\vec{b})$ (D) $\frac{1}{5}(-\vec{b} + 4\vec{a})$

Open solution

Q4. JEE Main 2026 (Online) 5th April Morning Shift

Let $\vec{a} = \sqrt{7}\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{j} + 2\hat{k}$. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} + \vec{a} \times \vec{b} = \vec{0}$ and $\vec{r} \cdot \vec{a} = 0$, then $|3\vec{r}|^2$ is equal to:

(A) 44 (B) 54 (C) 86 (D) 132

Open solution

Q5. JEE Main 2026 (Online) 4th April Evening Shift

Let $\hat{u}$ and $\hat{v}$ be unit vectors inclined at an acute angle such that $|\hat{u} \times \hat{v}| = \frac{\sqrt{3}}{2}$. If $\vec{A} = \lambda\hat{u} + \hat{v} + (\hat{u} \times \hat{v})$, then λ is equal to:

(A) $\frac{4}{3}(\vec{A} \cdot \hat{u}) - \frac{2}{3}(\vec{A} \cdot \hat{v})$

(B) $\frac{2}{3}(\vec{A} \cdot \hat{u}) - \frac{1}{3}(\vec{A} \cdot \hat{v})$

(C) $\frac{4}{3}(\vec{A} \cdot \hat{u}) + \frac{2}{3}(\vec{A} \cdot \hat{v})$

(D) $(\vec{A} \cdot \hat{u}) - \frac{1}{2}(\vec{A} \cdot \hat{v})$

Open solution

Q6. JEE Main 2026 (Online) 2nd April Evening Shift

Let the vectors $\vec{a} = -\hat{i} + \hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} + \hat{k}$. For some $\lambda, \mu \in \mathbb{R}$, let $\vec{c} = \lambda\vec{a} + \mu\vec{b}$.

If $\vec{c} \cdot (3\hat{i} - 6\hat{j} + 2\hat{k}) = 10$ and $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = -2$, then $|\vec{c}|^2$ is equal to :

(A) 8 (B) 12 (C) 14 (D) 15

Open solution

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Q7. JEE Main 2026 (Online) 2nd April Evening Shift

Two adjacent sides of a parallelogram PQRS are given by $\overrightarrow{PQ} = \hat{j} + \hat{k}$ and $\overrightarrow{PS} = \hat{i} - \hat{j}$. If the side PS is rotated about the point P by an acute angle α in the plane of the parallelogram so that it becomes perpendicular to the side PQ, then $\sin^2\left(\frac{5\alpha}{2}\right) - \sin^2\left(\frac{\alpha}{2}\right)$ is equal to:

(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{\sqrt{3}}{4}$ (D) $\frac{2\sqrt{3}}{5}$

Open solution

Q8. JEE Main 2026 (Online) 2nd April Morning Shift

If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = 2$ and $|\vec{b}| = 3$, then the maximum value of $3|(3\vec{a} + 2\vec{b})| + 4|(3\vec{a} - 2\vec{b})|$ is :

(A) 30 (B) 36 (C) 60 (D) 72

Open solution

Q9. JEE Main 2026 (Online) 28th January Evening Shift

Let P be a point in the plane of the vectors $\overrightarrow{AB} = 3\hat{i} + \hat{j} - \hat{k}$ and $\overrightarrow{AC} = \hat{i} - \hat{j} + 3\hat{k}$ such that P is equidistant from the lines AB and AC. If $|\overrightarrow{AP}| = \frac{\sqrt{5}}{2}$, then the area of the triangle ABP is :

(A) $\frac{3}{2}$ (B) $\frac{\sqrt{30}}{4}$ (C) $\frac{\sqrt{26}}{4}$ (D) 2

Open solution

Q10. JEE Main 2026 (Online) 28th January Morning Shift

For three unit vectors $\vec{a}, \vec{b}, \vec{c}$ satisfying

$|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9$ and $|2\vec{a} + k\vec{b} + k\vec{c}| = 3$,

the positive value of k is :

(A) 4 (B) 5 (C) 6 (D) 3

Open solution

Q11. JEE Main 2026 (Online) 24th January Evening Shift

Let $\vec{a} = 2\hat{i} - \hat{j} - \hat{k}, \vec{b} = \hat{i} + 3\hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} + \hat{j} + 3\hat{k}$. Let $\vec{v}$ be the vector in the plane of the vectors $\vec{a}$ and $\vec{b}$, such that the length of its projection on the vector $\vec{c}$ is $\frac{1}{\sqrt{14}}$. Then $|\vec{v}|$ is equal to

(A) $\frac{\sqrt{35}}{2}$ (B) 13 (C) $\frac{\sqrt{21}}{2}$ (D) 7

Open solution

Q12. JEE Main 2026 (Online) 24th January Evening Shift

Let $\vec{a} = 2\hat{i} - 5\hat{j} + 5\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + 3\hat{k}$. If $\vec{c}$ is a vector such that $2(\vec{a} \times \vec{c}) + 3(\vec{b} \times \vec{c}) = \vec{0}$ and $(\vec{a} - \vec{b}) \cdot \vec{c} = -97$, then $|\vec{c} \times \hat{k}|^2$ is equal to

(A) 193 (B) 205 (C) 218 (D) 233

Open solution

Q13. JEE Main 2026 (Online) 24th January Morning Shift

Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}, \vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \vec{a} \times \vec{b}$. Let $\vec{d}$ be a vector such that $|\vec{d} - \vec{a}| = \sqrt{11}, |\vec{c} \times \vec{d}| = 3$ and the angle between $\vec{c}$ and $\vec{d}$ is $\frac{\pi}{4}$. Then $\vec{a} \cdot \vec{d}$ is equal to

(A) 0 (B) 1 (C) 3 (D) 11

Open solution

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Q14. JEE Main 2026 (Online) 23rd January Evening Shift

Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $\vec{a} \times \vec{b} = 2(\vec{a} \times \vec{c})$. If $|\vec{a}| = 1, |\vec{b}| = 4, |\vec{c}| = 2$, and the angle between $\vec{b}$ and $\vec{c}$ is 60°, then $|\vec{a} \cdot \vec{c}|$ is equal to

(A) 2 (B) 0 (C) 1 (D) 4

Open solution

Q15. JEE Main 2026 (Online) 23rd January Evening Shift

Let $\vec{a} = \hat{\mathrm{i}} - 2\hat{\mathrm{j}} + 3\hat{\mathrm{k}}, \vec{b} = 2\hat{\mathrm{i}} + \hat{\mathrm{j}} - \hat{\mathrm{k}}, \vec{c} = \lambda\hat{\mathrm{i}} + \hat{\mathrm{j}} + \hat{\mathrm{k}}$ and $\vec{v} = \vec{a} \times \vec{b}$. If $\vec{v} \cdot \vec{c} = 11$ and the length of the projection of $\vec{b}$ on $\vec{c}$ is p, then $9p^2$ is equal to

(A) 9 (B) 6 (C) 4 (D) 12

Open solution

Q16. JEE Main 2026 (Online) 23rd January Morning Shift

Let $\overrightarrow{\mathrm{a}} = -\hat{i} + \hat{j} + 2\hat{k}, \overrightarrow{\mathrm{b}} = \hat{i} - \hat{j} - 3\hat{k}, \overrightarrow{\mathrm{c}} = \overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{b}}$ and $\overrightarrow{\mathrm{d}} = \overrightarrow{\mathrm{c}} \times \overrightarrow{\mathrm{a}}$. Then $(\vec{a} - \vec{b}) \cdot \vec{d}$ is equal to :

(A) 2 (B) -2 (C) 4 (D) -4

Open solution

Q17. JEE Main 2026 (Online) 22nd January Evening Shift

Let $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \lambda\hat{j} + 2\hat{k}, \lambda \in \boldsymbol{Z}$ be two vectors. Let $\vec{c} = \vec{a} \times \vec{b}$ and $\vec{d}$ be a vector of magnitude 2 in yz-plane. If $|\vec{c}| = \sqrt{53}$, then the maximum possible value of $(\vec{c} \cdot \vec{d})^2$ is equal to :

(A) 52 (B) 26 (C) 104 (D) 208

Open solution

Q18. JEE Main 2026 (Online) 22nd January Morning Shift

Let $\overrightarrow{\mathrm{AB}} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\overrightarrow{\mathrm{AD}} = \hat{i} + 2\hat{j} + \lambda\hat{k}, \lambda \in \mathbb{R}$. Let the projection of the vector $\vec{v} = \hat{i} + \hat{j} + \hat{k}$ on the diagonal $\overrightarrow{\mathrm{AC}}$ of the parallelogram ABCD be of length one unit. If α, β, where $\alpha > \beta$, be the roots of the equation $\lambda^2 x^2 - 6\lambda x + 5 = 0$, then $2\alpha - \beta$ is equal to

(A) 3 (B) 6 (C) 4 (D) 1

Open solution

Q19. JEE Main 2026 (Online) 21st January Evening Shift

For a triangle ABC, let $\vec{p} = \overrightarrow{BC}, \vec{q} = \overrightarrow{CA}$ and $\vec{r} = \overrightarrow{BA}$. If $|\vec{p}| = 2\sqrt{3}, |\vec{q}| = 2$ and $\cos\theta = \frac{1}{\sqrt{3}}$, where θ is the angle between $\vec{p}$ and $\vec{q}$, then $|\vec{p} \times (\vec{q} - 3\vec{r})|^2 + 3|\vec{r}|^2$ is equal to :

(A) 200 (B) 220 (C) 410 (D) 340

Open solution

Q20. JEE Main 2026 (Online) 21st January Morning Shift

Let $\overrightarrow{\mathrm{a}} = -\hat{i} + 2\hat{j} + 2\hat{k}, \overrightarrow{\mathrm{b}} = 8\hat{i} + 7\hat{j} - 3\hat{k}$ and $\overrightarrow{\mathrm{c}}$ be a vector such that $\overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{c}} = \overrightarrow{\mathrm{b}}$. If $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = 4$, then $|\vec{a} + \vec{c}|^2$ is equal to :

(A) 30 (B) 33 (C) 27 (D) 35

Open solution

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Q21. JEE Main 2026 (Online) 21st January Morning Shift

Let $\vec{c}$ and $\vec{d}$ be vectors such that $|\vec{c}+\vec{d}| = \sqrt{29}$ and $\vec{c} \times (2\hat{i}+3\hat{j}+4\hat{k}) = (2\hat{i}+3\hat{j}+4\hat{k}) \times \vec{d}$. If $\lambda_1, \lambda_2\ (\lambda_1 > \lambda_2)$ are the possible values of $(\vec{c}+\vec{d}) \cdot (-7\hat{i}+2\hat{j}+3\hat{k})$, then the equation $\mathrm{K}^2x^2 + (\mathrm{K}^2 - 5\,\mathrm{K} + \lambda_1)xy + \left(3\,\mathrm{K} + \frac{\lambda_2}{2}\right)y^2 - 8x + 12y + \lambda_2 = 0$ represents a circle, for K equal to :

Open solution

(A) 4 (B) -1 (C) 2 (D) 1

Q22. JEE Main 2025 (Online) 8th April Evening Shift

Let $\vec{a} = \hat{i}+2\hat{j}+\hat{k}$ and $\vec{b} = 2\hat{i}+\hat{j}-\hat{k}$. Let $\hat{c}$ be a unit vector in the plane of the vectors $\vec{a}$ and $\vec{b}$ and be perpendicular to $\vec{a}$. Then such a vector $\hat{c}$ is:

Open solution

(A) $\frac{1}{\sqrt{2}}(-\hat{i}+\hat{k})$ (B) $\frac{1}{\sqrt{5}}(\hat{j}-2\hat{k})$

(C) $\frac{1}{\sqrt{3}}(\hat{i}-\hat{j}+\hat{k})$ (D) $\frac{1}{\sqrt{3}}(-\hat{i}+\hat{j}-\hat{k})$

Q23. JEE Main 2025 (Online) 7th April Evening Shift

Let $\vec{a}$ and $\vec{b}$ be the vectors of the same magnitude such that $\frac{|\vec{a}+\vec{b}|+|\vec{a}-\vec{b}|}{|\vec{a}+\vec{b}|-|\vec{a}-\vec{b}|} = \sqrt{2}+1$. Then $\frac{|\vec{a}+\vec{b}|^2}{|\vec{a}|^2}$ is :

Open solution

(A) $2+\sqrt{2}$ (B) $2+4\sqrt{2}$ (C) $4+2\sqrt{2}$ (D) $1+\sqrt{2}$

Q24. JEE Main 2025 (Online) 7th April Morning Shift

Let the angle $\theta, 0 < \theta < \frac{\pi}{2}$ between two unit vectors $\hat{a}$ and $\hat{b}$ be $\sin^{-1}\left(\frac{\sqrt{65}}{9}\right)$. If the vector $\vec{c} = 3\hat{a} + 6\hat{b} + 9(\hat{a} \times \hat{b})$, then the value of $9(\vec{c} \cdot \hat{a}) - 3(\vec{c} \cdot \hat{b})$ is

Open solution

(A) 31 (B) 29 (C) 24 (D) 27

Q25. JEE Main 2025 (Online) 4th April Morning Shift

Consider two vectors $\vec{u} = 3\hat{i} - \hat{j}$ and $\vec{v} = 2\hat{i} + \hat{j} - \lambda\hat{k}, \lambda > 0$. The angle between them is given by $\cos^{-1}\left(\frac{\sqrt{5}}{2\sqrt{7}}\right)$. Let $\vec{v} = \vec{v}_1 + \vec{v_2}$, where $\vec{v}_1$ is parallel to $\vec{u}$ and $\vec{v_2}$ is perpendicular to $\vec{u}$. Then the value of $\left|\vec{v_1}\right|^2 + \left|\vec{v_2}\right|^2$ is equal to

Open solution

(A) $\frac{23}{2}$ (B) $\frac{25}{2}$ (C) 10 (D) 14

Q26. JEE Main 2025 (Online) 2nd April Evening Shift

Let $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}$, $\vec{b} = 3\hat{i} + 2\hat{j} + 5\hat{k}$ and a vector $\vec{c}$ be such that $(\vec{a} - \vec{c}) \times \vec{b} = -18\hat{i} - 3\hat{j} + 12\hat{k}$ and $\vec{a} \cdot \vec{c} = 3$. If $\vec{b} \times \vec{c} = \vec{d}$, then $|\vec{a} \cdot \vec{d}|$ is equal to :

Open solution

(A) 15 (B) 18 (C) 12 (D) 9

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Q27. JEE Main 2025 (Online) 2nd April Morning Shift

If $\vec{a}$ is a nonzero vector such that its projections on the vectors $2\hat{i}-\hat{j}+2\hat{k}, \hat{i}+2\hat{j}-2\hat{k}$ and $\hat{k}$ are equal, then a unit vector along $\vec{a}$ is :

(A) $\frac{1}{\sqrt{155}}(-7\hat{i}+9\hat{j}+5\hat{k})$ (B) $\frac{1}{\sqrt{155}}(-7\hat{i}+9\hat{j}-5\hat{k})$

(C) $\frac{1}{\sqrt{155}}(7\hat{i}+9\hat{j}-5\hat{k})$ (D) $\frac{1}{\sqrt{155}}(7\hat{i}+9\hat{j}+5\hat{k})$

Open solution

Q28. JEE Main 2025 (Online) 29th January Evening Shift

Let $\hat{a}$ be a unit vector perpendicular to the vectors $\vec{b}=\hat{i}-2\hat{j}+3\hat{k}$ and $\vec{c}=2\hat{i}+3\hat{j}-\hat{k}$, and $\hat{a}$ makes an angle of $\cos^{-1}\left(-\frac{1}{3}\right)$ with the vector $\hat{i}+\hat{j}+\hat{k}$. If $\hat{a}$ makes an angle of $\frac{\pi}{3}$ with the vector $\hat{i}+\alpha\hat{j}+\hat{k}$, then the value of α is:

(A) $\sqrt{3}$ (B) $\sqrt{6}$ (C) $-\sqrt{6}$ (D) $-\sqrt{3}$

Open solution

Q29. JEE Main 2025 (Online) 29th January Morning Shift

Let $\vec{a}=2\hat{i}-\hat{j}+3\hat{k}$, $\vec{b}=3\hat{i}-5\hat{j}+\hat{k}$ and $\vec{c}$ be a vector such that $\vec{a}\times\vec{c}=\vec{c}\times\vec{b}$ and $(\vec{a}+\vec{c})\cdot(\vec{b}+\vec{c})=168$. Then the maximum value of $|\vec{c}|^2$ is :

(A) 77 (B) 154 (C) 308 (D) 462

Open solution

Q30. JEE Main 2025 (Online) 28th January Evening Shift

If the components of $\vec{a}=\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$ along and perpendicular to $\vec{b}=3\hat{i}+\hat{j}-\hat{k}$ respectively, are $\frac{16}{11}(3\hat{i}+\hat{j}-\hat{k})$ and $\frac{1}{11}(-4\hat{i}-5\hat{j}-17\hat{k})$, then $\alpha^2+\beta^2+\gamma^2$ is equal to :

(A) 16 (B) 23 (C) 26 (D) 18

Open solution

Q31. JEE Main 2025 (Online) 28th January Evening Shift

Let A, B, C be three points in xy-plane, whose position vectors are given by $\sqrt{3}\hat{i}+\hat{j}, \hat{i}+\sqrt{3}\hat{j}$ and $a\hat{i}+(1-a)\hat{j}$ respectively with respect to the origin O . If the distance of the point C from the line bisecting the angle between the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ is $\frac{9}{\sqrt{2}}$, then the sum of all the possible values of a is :

(A) 2 (B) 0 (C) $\frac{9}{2}$ (D) 1

Open solution

Q32. JEE Main 2025 (Online) 24th January Evening Shift

Let the position vectors of three vertices of a triangle be $4\vec{p}+\vec{q}-3\vec{r}$, $-5\vec{p}+\vec{q}+2\vec{r}$ and $2\vec{p}-\vec{q}+2\vec{r}$. If the position vectors of the orthocenter and the circumcenter of the triangle are $\frac{\vec{p}+\vec{q}+\vec{r}}{4}$ and $\alpha\vec{p}+\beta\vec{q}+\gamma\vec{r}$ respectively, then $\alpha+2\beta+5\gamma$ is equal to :

(A) 4 (B) 3 (C) 1 (D) 6

Open solution

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Q33. JEE Main 2025 (Online) 24th January Evening Shift

Let $\vec{a} = 3\hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = \vec{a} \times (\hat{i} - 2\hat{k})$ and $\vec{c} = \vec{b} \times \hat{k}$. Then the projection of $\vec{c} - 2\hat{j}$ on $\vec{a}$ is :

(A) $2\sqrt{7}$ (B) $3\sqrt{7}$ (C) $\sqrt{14}$ (D) $2\sqrt{14}$

Open solution

Q34. JEE Main 2025 (Online) 24th January Morning Shift

Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{b} = 3\hat{i} + \hat{j} - \hat{k}$ and $\vec{c}$ be three vectors such that $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$. If the vector $\vec{c}$ is perpendicular to $\vec{b}$ and $\vec{a} \cdot \vec{c} = 5$, then $|\vec{c}|$ is equal to

(A) $\sqrt{\frac{11}{6}}$ (B) $\frac{1}{3\sqrt{2}}$ (C) 18 (D) 16

Open solution

Q35. JEE Main 2025 (Online) 23rd January Evening Shift

Let the point A divide the line segment joining the points $\mathrm{P}(-1, -1, 2)$ and $\mathrm{Q}(5, 5, 10)$ internally in the ratio $r : 1 (r > 0)$. If O is the origin and $(\overrightarrow{\mathrm{OQ}} \cdot \overrightarrow{\mathrm{OA}}) - \frac{1}{5}|\overrightarrow{\mathrm{OP}} \times \overrightarrow{\mathrm{OA}}|^2 = 10$, then the value of r is :

(A) $\sqrt{7}$ (B) 14 (C) 7 (D) 3

Open solution

Q36. JEE Main 2025 (Online) 23rd January Morning Shift

Let the position vectors of the vertices A, B and C of a tetrahedron ABCD be $\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathrm{k}}, \hat{\mathrm{i}} + 3\hat{\mathrm{j}} - 2\hat{\mathrm{k}}$ and $2\hat{i} + \hat{j} - \hat{k}$ respectively. The altitude from the vertex D to the opposite face ABC meets the median line segment through A of the triangle ABC at the point E. If the length of AD is $\frac{\sqrt{110}}{3}$ and the volume of the tetrahedron is $\frac{\sqrt{805}}{6\sqrt{2}}$, then the position vector of E is

(A) $\frac{1}{6}(7\hat{\mathrm{i}} + 12\hat{\mathrm{j}} + \hat{\mathrm{k}})$ (B) $\frac{1}{12}(7\hat{\mathrm{i}} + 4\hat{\mathrm{j}} + 3\hat{\mathrm{k}})$

(C) $\frac{1}{6}(12\hat{i} + 12\hat{j} + \hat{k})$ (D) $\frac{1}{2}(\hat{i} + 4\hat{j} + 7\hat{k})$

Open solution

Q37. JEE Main 2025 (Online) 23rd January Morning Shift

Let the arc AC of a circle subtend a right angle at the centre O. If the point B on the arc AC, divides the arc AC such that $\frac{\text{length of arc } AB}{\text{length of arc } BC} = \frac{1}{5}$, and $\overrightarrow{OC} = \alpha\overrightarrow{OA} + \beta\overrightarrow{OB}$, then $\alpha + \sqrt{2}(\sqrt{3} - 1)\beta$ is equal to

(A) $2\sqrt{3}$ (B) $5\sqrt{3}$ (C) $2 + \sqrt{3}$ (D) $2 - \sqrt{3}$

Open solution

Q38. JEE Main 2025 (Online) 22nd January Evening Shift

Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that the angle between them is $\frac{\pi}{3}$. If $\lambda\vec{a} + 2\vec{b}$ and $3\vec{a} - \lambda\vec{b}$ are perpendicular to each other, then the number of values of λ in $[-1, 3]$ is :

(A) 1 (B) 3 (C) 2 (D) 0

Open solution

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Q39. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

Between the following two statements:

Statement I : Let $\vec{a}=\hat{i}+2\hat{j}-3\hat{k}$ and $\vec{b}=2\hat{i}+\hat{j}-\hat{k}$. Then the vector $\vec{r}$ satisfying $\vec{a}\times\vec{r}=\vec{a}\times\vec{b}$ and $\vec{a}\cdot\vec{r}=0$ is of magnitude $\sqrt{10}$.

Statement II : In a triangle ABC, $\cos 2A+\cos 2B+\cos 2C\geq -\frac{3}{2}$.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Q40. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

Let $\vec{a}=2\hat{i}+\alpha\hat{j}+\hat{k},\vec{b}=-\hat{i}+\hat{k},\vec{c}=\beta\hat{j}-\hat{k}$, where α and β are integers and $\alpha\beta=-6$. Let the values of the ordered pair (α,β), for which the area of the parallelogram of diagonals $\vec{a}+\vec{b}$ and $\vec{b}+\vec{c}$ is $\frac{\sqrt{21}}{2}$, be (α_1,β_1) and (α_2,β_2). Then $\alpha_1^2+\beta_1^2-\alpha_2\beta_2$ is equal to

(A) 21 (B) 24 (C) 19 (D) 17

Q41. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

Let three vectors $\overrightarrow{a}=\alpha\hat{i}+4\hat{j}+2\hat{k},\overrightarrow{b}=5\hat{i}+3\hat{j}+4\hat{k},\overrightarrow{c}=x\hat{i}+y\hat{j}+z\hat{k}$ form a triangle such that $\vec{c}=\vec{a}-\vec{b}$ and the area of the triangle is $5\sqrt{6}$. If α is a positive real number, then $|\vec{c}|^2$ is equal to:

(A) 14 (B) 12 (C) 16 (D) 10

Q42. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

Let $\overrightarrow{OA}=2\vec{a},\overrightarrow{OB}=6\vec{a}+5\vec{b}$ and $\overrightarrow{OC}=3\vec{b}$, where O is the origin. If the area of the parallelogram with adjacent sides $\overrightarrow{OA}$ and $\overrightarrow{OC}$ is 15 sq. units, then the area (in sq. units) of the quadrilateral $OABC$ is equal to:

(A) 32 (B) 38 (C) 35 (D) 40

Q43. JEE Main 2024 (Online) 8th April Evening Shift

Open solution

Let $\overrightarrow{a}=4\hat{i}-\hat{j}+\hat{k},\overrightarrow{b}=11\hat{i}-\hat{j}+\hat{k}$ and $\overrightarrow{c}$ be a vector such that $(\overrightarrow{a}+\overrightarrow{b})\times\overrightarrow{c}=\overrightarrow{c}\times(-2\overrightarrow{a}+3\overrightarrow{b})$. If $(2\vec{a}+3\vec{b})\cdot\vec{c}=1670$, then $|\vec{c}|^2$ is equal to:

(A) 1600 (B) 1618 (C) 1627 (D) 1609

Q44. JEE Main 2024 (Online) 8th April Evening Shift

Open solution

Let $\overrightarrow{a}=\hat{i}+2\hat{j}+3\hat{k},\overrightarrow{b}=2\hat{i}+3\hat{j}-5\hat{k}$ and $\overrightarrow{c}=3\hat{i}-\hat{j}+\lambda\hat{k}$ be three vectors. Let $\overrightarrow{r}$ be a unit vector along $\vec{b}+\vec{c}$. If $\vec{r}\cdot\vec{a}=3$, then 3λ is equal to:

(A) 21 (B) 25 (C) 27 (D) 30

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Q45. JEE Main 2024 (Online) 8th April Morning Shift

The set of all α, for which the vectors $\vec{a} = \alpha t\hat{i} + 6\hat{j} - 3\hat{k}$ and $\vec{b} = t\hat{i} - 2\hat{j} - 2\alpha t\hat{k}$ are inclined at an obtuse angle for all $t \in \mathbb{R}$, is

(A) $[0, 1)$ (B) $\left(-\frac{4}{3}, 0\right]$ (C) $(-2, 0]$ (D) $\left(-\frac{4}{3}, 1\right)$

Open solution

Q46. JEE Main 2024 (Online) 6th April Evening Shift

Let $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}, \vec{b} = ((\vec{a} \times (\hat{i} + \hat{j})) \times \hat{i}) \times \hat{i}$. Then the square of the projection of $\vec{a}$ on $\vec{b}$ is:

(A) $\frac{1}{3}$ (B) $\frac{1}{5}$ (C) 2 (D) $\frac{2}{3}$

Open solution

Q47. JEE Main 2024 (Online) 6th April Evening Shift

Let $\overrightarrow{a} = 6\hat{i} + \hat{j} - \hat{k}$ and $\overrightarrow{b} = \hat{i} + \hat{j}$. If $\overrightarrow{c}$ is a vector such that $|\overrightarrow{c}| \geq 6, \overrightarrow{a} \cdot \overrightarrow{c} = 6|\overrightarrow{c}|, |\overrightarrow{c} - \overrightarrow{a}| = 2\sqrt{2}$ and the angle between $\vec{a} \times \vec{b}$ and $\vec{c}$ is 60°, then $|(\vec{a} \times \vec{b}) \times \vec{c}|$ is equal to:

(A) $\frac{3}{2}\sqrt{6}$ (B) $\frac{9}{2}(6 - \sqrt{6})$ (C) $\frac{9}{2}(6 + \sqrt{6})$ (D) $\frac{3}{2}\sqrt{3}$

Open solution

Q48. JEE Main 2024 (Online) 6th April Morning Shift

If $A(3, 1, -1), B\left(\frac{5}{3}, \frac{7}{3}, \frac{1}{3}\right), C(2, 2, 1)$ and $D\left(\frac{10}{3}, \frac{2}{3}, \frac{-1}{3}\right)$ are the vertices of a quadrilateral $ABCD$, then its area is

(A) $\frac{4\sqrt{2}}{3}$ (B) $\frac{2\sqrt{2}}{3}$ (C) $\frac{5\sqrt{2}}{3}$ (D) $2\sqrt{2}$

Open solution

Q49. JEE Main 2024 (Online) 5th April Evening Shift

Let $\vec{a} = 2\hat{i} + 5\hat{j} - \hat{k}, \vec{b} = 2\hat{i} - 2\hat{j} + 2\hat{k}$ and $\vec{c}$ be three vectors such that $(\vec{c} + \hat{i}) \times (\vec{a} + \vec{b} + \hat{i}) = \vec{a} \times (\vec{c} + \hat{i})$. If $\vec{a} \cdot \vec{c} = -29$, then $\vec{c} \cdot (-2\hat{i} + \hat{j} + \hat{k})$ is equal to:

(A) 15 (B) 10 (C) 5 (D) 12

Open solution

Q50. JEE Main 2024 (Online) 5th April Evening Shift

Consider three vectors $\vec{a}, \vec{b}, \vec{c}$. Let $|\vec{a}| = 2, |\vec{b}| = 3$ and $\vec{a} = \vec{b} \times \vec{c}$. If $\alpha \in \left[0, \frac{\pi}{3}\right]$ is the angle between the vectors $\vec{b}$ and $\vec{c}$, then the minimum value of $27|\vec{c} - \vec{a}|^2$ is equal to:

(A) 124 (B) 110 (C) 121 (D) 105

Open solution

Q51. JEE Main 2024 (Online) 5th April Morning Shift

If $A(1, -1, 2), B(5, 7, -6), C(3, 4, -10)$ and $D(-1, -4, -2)$ are the vertices of a quadrilateral ABCD, then its area is :

(A) $24\sqrt{7}$ (B) $48\sqrt{7}$ (C) $24\sqrt{29}$ (D) $12\sqrt{29}$

Open solution

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Q52. JEE Main 2024 (Online) 4th April Evening Shift

For $\lambda > 0$, let θ be the angle between the vectors $\vec{a} = \hat{i} + \lambda\hat{j} - 3\hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$. If the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are mutually perpendicular, then the value of $(14\cos\theta)^2$ is equal to

(A) 25 (B) 50 (C) 20 (D) 40

Open solution

Q53. JEE Main 2024 (Online) 4th April Evening Shift

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{c} = x\hat{i} + 2\hat{j} + 3\hat{k}, x \in \mathbb{R}$. If $\vec{d}$ is the unit vector in the direction of $\vec{b} + \vec{c}$ such that $\vec{a} \cdot \vec{d} = 1$, then $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is equal to

(A) 3 (B) 9 (C) 11 (D) 6

Open solution

Q54. JEE Main 2024 (Online) 4th April Morning Shift

Let a unit vector which makes an angle of 60° with $2\hat{i} + 2\hat{j} - \hat{k}$ and an angle of 45° with $\hat{i} - \hat{k}$ be $\vec{C}$. Then $\vec{C} + \left(-\frac{1}{2}\hat{i} + \frac{1}{3\sqrt{2}}\hat{j} - \frac{\sqrt{2}}{3}\hat{k}\right)$ is:

(A) $-\frac{\sqrt{2}}{3}\hat{i} + \frac{\sqrt{2}}{3}\hat{j} + \left(\frac{1}{2} + \frac{2\sqrt{2}}{3}\right)\hat{k}$

(B) $\left(\frac{1}{\sqrt{3}} + \frac{1}{2}\right)\hat{i} + \left(\frac{1}{\sqrt{3}} - \frac{1}{3\sqrt{2}}\right)\hat{j} + \left(\frac{1}{\sqrt{3}} + \frac{\sqrt{2}}{3}\right)\hat{k}$

(C) $\frac{\sqrt{2}}{3}\hat{i} - \frac{1}{2}\hat{k}$

(D) $\frac{\sqrt{2}}{3}\hat{i} + \frac{1}{3\sqrt{2}}\hat{j} - \frac{1}{2}\hat{k}$

Open solution

Q55. JEE Main 2024 (Online) 1st February Evening Shift

Consider a $\triangle ABC$ where $A(1, 3, 2), B(-2, 8, 0)$ and $C(3, 6, 7)$. If the angle bisector of $\angle BAC$ meets the line BC at D, then the length of the projection of the vector $\overrightarrow{AD}$ on the vector $\overrightarrow{AC}$ is :

(A) $\frac{37}{2\sqrt{38}}$ (B) $\sqrt{19}$ (C) $\frac{39}{2\sqrt{38}}$ (D) $\frac{\sqrt{38}}{2}$

Open solution

Q56. JEE Main 2024 (Online) 1st February Morning Shift

Let $\overrightarrow{a} = -5\hat{i} + \hat{j} - 3\hat{k}, \overrightarrow{b} = \hat{i} + 2\hat{j} - 4\hat{k}$ and

$\overrightarrow{c} = (((\overrightarrow{a} \times \overrightarrow{b}) \times \hat{i}) \times \hat{i}) \times \hat{i}$. Then $\vec{c} \cdot (-\hat{i} + \hat{j} + \hat{k})$ is equal to :

(A) -12 (B) -10 (C) -13 (D) -15

Open solution

Q57. JEE Main 2024 (Online) 31st January Morning Shift

Let $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}, \vec{b} = 4\hat{i} + \hat{j} + 7\hat{k}$ and $\vec{c} = \hat{i} - 3\hat{j} + 4\hat{k}$ be three vectors. If a vector $\vec{p}$ satisfies $\vec{p} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{p} \cdot \vec{a} = 0$, then $\vec{p} \cdot (\hat{i} - \hat{j} - \hat{k})$ is equal to

(A) 24 (B) 32 (C) 36 (D) 28

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Q58. JEE Main 2024 (Online) 30th January Evening Shift

Let $\vec{a} = \hat{i} + \alpha\hat{j} + \beta\hat{k}, \alpha, \beta \in \mathbb{R}$. Let a vector $\vec{b}$ be such that the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$ and $|\vec{b}|^2 = 6$. If $\vec{a} \cdot \vec{b} = 3\sqrt{2}$, then the value of $(\alpha^2 + \beta^2)|\vec{a} \times \vec{b}|^2$ is equal to

(A) 85 (B) 90 (C) 75 (D) 95

Open solution

Q59. JEE Main 2024 (Online) 30th January Evening Shift

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{b}| = 1$ and $|\vec{b} \times \vec{a}| = 2$. Then $|(\vec{b} \times \vec{a}) - \vec{b}|^2$ is equal to

(A) 1 (B) 3 (C) 5 (D) 4

Open solution

Q60. JEE Main 2024 (Online) 30th January Morning Shift

Let $\overrightarrow{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\overrightarrow{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ be two vectors such that $|\overrightarrow{a}| = 1, \vec{a} \cdot \vec{b} = 2$ and $|\vec{b}| = 4$. If $\vec{c} = 2(\vec{a} \times \vec{b}) - 3\vec{b}$, then the angle between $\vec{b}$ and $\vec{c}$ is equal to:

(A) $\cos^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ (B) $\cos^{-1}\left(\frac{2}{3}\right)$

(C) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (D) $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$

Open solution

Q61. JEE Main 2024 (Online) 30th January Morning Shift

Let $A(2, 3, 5)$ and $C(-3, 4, -2)$ be opposite vertices of a parallelogram $ABCD$. If the diagonal $\overrightarrow{BD} = \hat{i} + 2\hat{j} + 3\hat{k}$, then the area of the parallelogram is equal to :

(A) $\frac{1}{2}\sqrt{410}$ (B) $\frac{1}{2}\sqrt{306}$ (C) $\frac{1}{2}\sqrt{586}$ (D) $\frac{1}{2}\sqrt{474}$

Open solution

Q62. JEE Main 2024 (Online) 29th January Evening Shift

Let a unit vector $\hat{u} = x\hat{i} + y\hat{j} + z\hat{k}$ make angles $\frac{\pi}{2}, \frac{\pi}{3}$ and $\frac{2\pi}{3}$ with the vectors $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k}, \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$ and $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$ respectively. If $\vec{v} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$ then $|\hat{u} - \vec{v}|^2$ is equal to

(A) $\frac{11}{2}$ (B) $\frac{5}{2}$ (C) 7 (D) 9

Open solution

Q63. JEE Main 2024 (Online) 29th January Evening Shift

Let $\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = 12\vec{a} + 4\vec{b}$ and $\overrightarrow{OC} = \vec{b}$, where O is the origin. If S is the parallelogram with adjacent sides OA and OC, then $\frac{\text{area of the quadrilateral OA BC}}{\text{area of S}}$ is equal to _____.

(A) 7 (B) 6 (C) 8 (D) 10

Open solution

Q64. JEE Main 2024 (Online) 29th January Morning Shift

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-zero vectors such that $\vec{b}$ and $\vec{c}$ are non-collinear. If $\vec{a} + 5\vec{b}$ is collinear with $\vec{c}, \vec{b} + 6\vec{c}$ is collinear with $\vec{a}$ and $\vec{a} + \alpha\vec{b} + \beta\vec{c} = \overrightarrow{0}$, then $\alpha + \beta$ is equal to

(A) 30 (B) -30 (C) -25 (D) 35

Open solution

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Q65. JEE Main 2024 (Online) 29th January Morning Shift

Let O be the origin and the position vectors of A and B be $2\hat{i}+2\hat{j}+\hat{k}$ and $2\hat{i}+4\hat{j}+4\hat{k}$ respectively. If the internal bisector of $\angle AOB$ meets the line AB at C, then the length of OC is

(A) $\frac{3}{2}\sqrt{34}$ (B) $\frac{2}{3}\sqrt{31}$ (C) $\frac{2}{3}\sqrt{34}$ (D) $\frac{3}{2}\sqrt{31}$

Open solution

Q66. JEE Main 2024 (Online) 27th January Evening Shift

Let the position vectors of the vertices A, B and C of a triangle be $2\hat{i}+2\hat{j}+\hat{k}, \hat{i}+2\hat{j}+2\hat{k}$ and $2\hat{i}+\hat{j}+2\hat{k}$ respectively. Let l_1, l_2 and l_3 be the lengths of perpendiculars drawn from the orthocentre of the triangle on the sides AB, BC and CA respectively, then $l_1^2+l_2^2+l_3^2$ equals:

(A) $\frac{1}{4}$ (B) $\frac{1}{5}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

Open solution

Q67. JEE Main 2024 (Online) 27th January Evening Shift

The position vectors of the vertices A, B and C of a triangle are $2\hat{i}-3\hat{j}+3\hat{k}, 2\hat{i}+2\hat{j}+3\hat{k}$ and $-\hat{i}+\hat{j}+3\hat{k}$ respectively. Let l denote the length of the angle bisector AD of $\angle BAC$ where D is on the line segment BC, then $2l^2$ equals :

(A) 45 (B) 50 (C) 42 (D) 49

Open solution

Q68. JEE Main 2024 (Online) 27th January Morning Shift

Let $\overrightarrow{a}=\hat{i}+2\hat{j}+\hat{k}$,
$\overrightarrow{b}=3(\hat{i}-\hat{j}+\hat{k})$.
Let $\overrightarrow{c}$ be the vector such that $\overrightarrow{a}\times\overrightarrow{c}=\overrightarrow{b}$ and $\vec{a}\cdot\vec{c}=3$.
Then $\vec{a}\cdot((\vec{c}\times\vec{b})-\vec{b}-\vec{c})$ is equal to :

(A) 32 (B) 36 (C) 24 (D) 20

Open solution

Q69. JEE Main 2023 (Online) 15th April Morning Shift

Let ABCD be a quadrilateral. If E and F are the mid points of the diagonals AC and BD respectively and $(\overrightarrow{AB}-\overrightarrow{BC})+(\overrightarrow{AD}-\overrightarrow{DC})=k\overrightarrow{FE}$, then k is equal to :

(A) -2 (B) 4 (C) -4 (D) 2

Open solution

Q70. JEE Main 2023 (Online) 13th April Evening Shift

Let $|\vec{a}|=2, |\vec{b}|=3$ and the angle between the vectors $\vec{a}$ and $\vec{b}$ be $\frac{\pi}{4}$. Then $|(\vec{a}+2\vec{b})\times(2\vec{a}-3\vec{b})|^2$ is equal to :

(A) 441 (B) 482 (C) 841 (D) 882

Open solution

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Q71. JEE Main 2023 (Online) 13th April Evening Shift

Open solution

Let for a triangle ABC,

$\overrightarrow{\mathrm{AB}} = -2\hat{i} + \hat{j} + 3\hat{k}$

$\overrightarrow{\mathrm{CB}} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$

$\overrightarrow{\mathrm{CA}} = 4\hat{i} + 3\hat{j} + \delta\hat{k}$

If $\delta > 0$ and the area of the triangle ABC is $5\sqrt{6}$, then $\overrightarrow{CB} \cdot \overrightarrow{CA}$ is equal to

(A) 60 (B) 54 (C) 120 (D) 108

Q72. JEE Main 2023 (Online) 13th April Morning Shift

Open solution

Let $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}, \vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. If a vector $\vec{d}$ satisfies $\vec{d} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{d} \cdot \vec{a} = 24$, then $|\vec{d}|^2$ is equal to :

(A) 313 (B) 413 (C) 423 (D) 323

Q73. JEE Main 2023 (Online) 11th April Morning Shift

Open solution

For any vector $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, with $10\,|a_i| < 1, i = 1, 2, 3$, consider the following statements :

(A): $\max\{|a_1|, |a_2|, |a_3|\} \leq |\vec{a}|$

(B) : $|\vec{a}| \leq 3\max\{|a_1|, |a_2|, |a_3|\}$

(A) Only (B) is true (B) Only (A) is true

(C) Neither (A) nor (B) is true (D) Both (A) and (B) are true

Q74. JEE Main 2023 (Online) 11th April Morning Shift

Open solution

Let $\vec{a}$ be a non-zero vector parallel to the line of intersection of the two planes described by $\hat{i} + \hat{j}, \hat{i} + \hat{k}$ and $\hat{i} - \hat{j}, \hat{j} - \hat{k}$. If θ is the angle between the vector $\vec{a}$ and the vector $\vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$ and $\vec{a} \cdot \vec{b} = 6$, then the ordered pair $(\theta, |\vec{a} \times \vec{b}|)$ is equal to :

(A) $\left(\frac{\pi}{3}, 3\sqrt{6}\right)$ (B) $\left(\frac{\pi}{3}, 6\right)$

(C) $\left(\frac{\pi}{4}, 3\sqrt{6}\right)$ (D) $\left(\frac{\pi}{4}, 6\right)$

Q75. JEE Main 2023 (Online) 10th April Evening Shift

Open solution

Let $\vec{a} = 2\hat{i} + 7\hat{j} - \hat{k}, \vec{b} = 3\hat{i} + 5\hat{k}$ and $\vec{c} = \hat{i} - \hat{j} + 2\hat{k}$. Let $\vec{d}$ be a vector which is perpendicular to both $\vec{a}$ and $\vec{b}$, and $\vec{c} \cdot \vec{d} = 12$. Then $(-\hat{i} + \hat{j} - \hat{k}) \cdot (\vec{c} \times \vec{d})$ is equal to :

(A) 24 (B) 42 (C) 44 (D) 48

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Q76. JEE Main 2023 (Online) 10th April Evening Shift

If the points P and Q are respectively the circumcenter and the orthocentre of a $\triangle \mathrm{ABC}$, then $\overrightarrow{\mathrm{PA}}+\overrightarrow{\mathrm{PB}}+\overrightarrow{\mathrm{PC}}$ is equal to :

(A) $\overrightarrow{QP}$ (B) $\overrightarrow{PQ}$ (C) $2\overrightarrow{PQ}$ (D) $2\overrightarrow{QP}$

Open solution

Q77. JEE Main 2023 (Online) 10th April Morning Shift

Let O be the origin and the position vector of the point P be $-\hat{i}-2\hat{j}+3\hat{k}$. If the position vectors of the points A, B and C are $-2\hat{i}+\hat{j}-3\hat{k}, 2\hat{i}+4\hat{j}-2\hat{k}$ and $-4\hat{i}+2\hat{j}-\hat{k}$ respectively, then the projection of the vector $\overrightarrow{OP}$ on a vector perpendicular to the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ is :

(A) $\frac{7}{3}$ (B) 3 (C) $\frac{10}{3}$ (D) $\frac{8}{3}$

Open solution

Q78. JEE Main 2023 (Online) 10th April Morning Shift

An arc PQ of a circle subtends a right angle at its centre O. The mid point of the arc PQ is R. If $\overrightarrow{OP}=\vec{u}, \overrightarrow{OR}=\vec{v}$, and $\overrightarrow{OQ}=\alpha\vec{u}+\beta\vec{v}$, then α, β^2 are the roots of the equation :

(A) $x^2+x-2=0$ (B) $3x^2+2x-1=0$ (C) $3x^2-2x-1=0$ (D) $x^2-x-2=0$

Open solution

Q79. JEE Main 2023 (Online) 8th April Evening Shift

The area of the quadrilateral ABCD with vertices $\mathrm{A}(2,1,1), \mathrm{B}(1,2,5), \mathrm{C}(-2,-3,5)$ and $\mathrm{D}(1,-6,-7)$ is equal to :

(A) 48 (B) $8\sqrt{38}$ (C) 54 (D) $9\sqrt{38}$

Open solution

Q80. JEE Main 2023 (Online) 8th April Morning Shift

If the points with position vectors $\alpha\hat{i}+10\hat{j}+13\hat{k}, 6\hat{i}+11\hat{j}+11\hat{k}, \frac{9}{2}\hat{i}+\beta\hat{j}-8\hat{k}$ are collinear, then $(19\alpha-6\beta)^2$ is equal to :

(A) 16 (B) 49 (C) 36 (D) 25

Open solution

Q81. JEE Main 2023 (Online) 6th April Morning Shift

Let $\vec{a}=2\hat{i}+3\hat{j}+4\hat{k}, \vec{b}=\hat{i}-2\hat{j}-2\hat{k}$ and $\vec{c}=-\hat{i}+4\hat{j}+3\hat{k}$. If $\vec{d}$ is a vector perpendicular to both $\vec{b}$ and $\vec{c}$, and $\vec{a}\cdot\vec{d}=18$, then $|\vec{a}\times\vec{d}|^2$ is equal to :

(A) 680 (B) 720 (C) 760 (D) 640

Open solution

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Q82. JEE Main 2023 (Online) 1st February Evening Shift

Let $\vec{a} = 5\hat{i} - \hat{j} - 3\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} + 5\hat{k}$ be two vectors. Then which one of the following statements is TRUE ?

(A) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{-13}{\sqrt{35}}$ and the direction of the projection vector is opposite to the direction of $\vec{b}$.

(B) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{13}{\sqrt{35}}$ and the direction of the projection vector is opposite to the direction of $\vec{b}$.

(C) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{13}{\sqrt{35}}$ and the direction of the projection vector is same as of $\vec{b}$.

(D) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{-13}{\sqrt{35}}$ and the direction of the projection vector is same as of $\vec{b}$.

Open solution

Q83. JEE Main 2023 (Online) 1st February Evening Shift

Let $\vec{a} = 2\hat{i} - 7\hat{j} + 5\hat{k}, \vec{b} = \hat{i} + \hat{k}$ and $\vec{c} = \hat{i} + 2\hat{j} - 3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = \vec{c} \times \vec{a}$ and $\vec{r} \cdot \vec{b} = 0$, then $|\vec{r}|$ is equal to :

(A) $\frac{11}{7}$ (B) $\frac{11}{5}\sqrt{2}$ (C) $\frac{\sqrt{914}}{7}$ (D) $\frac{11}{7}\sqrt{2}$

Open solution

Q84. JEE Main 2023 (Online) 31st January Evening Shift

Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{b} = \hat{i} - \hat{j} + 2\hat{k}$ and $\vec{c} = 5\hat{i} - 3\hat{j} + 3\hat{k}$ be three vectors. If $\vec{r}$ is a vector such that, $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$, then $25|\vec{r}|^2$ is equal to :

(A) 336 (B) 449 (C) 339 (D) 560

Open solution

Q85. JEE Main 2023 (Online) 31st January Morning Shift

Let $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$, and $\vec{b}$ and $\vec{c}$ be two nonzero vectors such that $|\vec{a} + \vec{b} + \vec{c}| = |\vec{a} + \vec{b} - \vec{c}|$ and $\vec{b} \cdot \vec{c} = 0$. Consider the following two statements:

(A) $|\vec{a} + \lambda\vec{c}| \geq |\vec{a}|$ for all $\lambda \in \mathbb{R}$.

(B) $\vec{a}$ and $\vec{c}$ are always parallel.

Then,

(A) only (B) is correct

(B) both (A) and (B) are correct

(C) only (A) is correct

(D) neither (A) nor (B) is correct

Open solution

Q86. JEE Main 2023 (Online) 30th January Evening Shift

Let $\vec{a}$ and $\vec{b}$ be two vectors, Let $|\vec{a}| = 1, |\vec{b}| = 4$ and $\vec{a} \cdot \vec{b} = 2$. If $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$, then the value of $\vec{b} \cdot \vec{c}$ is :

(A) -48 (B) -60 (C) -84 (D) -24

Open solution

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Q87. JEE Main 2023 (Online) 30th January Morning Shift

Let a unit vector $\widehat{OP}$ make angles α, β, γ with the positive directions of the co-ordinate axes OX, OY, OZ respectively, where $\beta \in \left(0, \frac{\pi}{2}\right)$. If $\widehat{OP}$ is perpendicular to the plane through points $(1,2,3), (2,3,4)$ and $(1,5,7)$, then which one of the following is true?

(A) $\alpha \in \left(\frac{\pi}{2}, \pi\right)$ and $\gamma \in \left(\frac{\pi}{2}, \pi\right)$

(B) $\alpha \in \left(0, \frac{\pi}{2}\right)$ and $\gamma \in \left(\frac{\pi}{2}, \pi\right)$

(C) $\alpha \in \left(\frac{\pi}{2}, \pi\right)$ and $\gamma \in \left(0, \frac{\pi}{2}\right)$

(D) $\alpha \in \left(0, \frac{\pi}{2}\right)$ and $\gamma \in \left(0, \frac{\pi}{2}\right)$

Open solution

Q88. JEE Main 2023 (Online) 29th January Evening Shift

If $\vec{a} = \hat{i} + 2\hat{k}, \vec{b} = \hat{i} + \hat{j} + \hat{k}, \vec{c} = 7\hat{i} - 3\hat{j} + 4\hat{k}, \vec{r} \times \vec{b} + \vec{b} \times \vec{c} = \vec{0}$ and $\vec{r} \cdot \vec{a} = 0$. Then $\vec{r} \cdot \vec{c}$ is equal to :

(A) 36 (B) 30 (C) 34 (D) 32

Open solution

Q89. JEE Main 2023 (Online) 29th January Evening Shift

Let $\vec{a} = 4\hat{i} + 3\hat{j}$ and $\vec{b} = 3\hat{i} - 4\hat{j} + 5\hat{k}$. If $\vec{c}$ is a vector such that $\vec{c} \cdot \left(\vec{a} \times \vec{b}\right) + 25 = 0, \vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = 4$, and projection of $\vec{c}$ on $\vec{a}$ is 1, then the projection of $\vec{c}$ on $\vec{b}$ equals :

(A) $\frac{3}{\sqrt{2}}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{5}$ (D) $\frac{5}{\sqrt{2}}$

Open solution

Q90. JEE Main 2023 (Online) 29th January Morning Shift

If the vectors $\vec{a} = \lambda\hat{i} + \mu\hat{j} + 4\hat{k}, \vec{b} = -2\hat{i} + 4\hat{j} - 2\hat{k}$ and $\vec{c} = 2\hat{i} + 3\hat{j} + \hat{k}$ are coplanar and the projection of $\vec{a}$ on the vector $\vec{b}$ is $\sqrt{54}$ units, then the sum of all possible values of $\lambda + \mu$ is equal to :

(A) 24 (B) 0 (C) 18 (D) 6

Open solution

Q91. JEE Main 2023 (Online) 25th January Morning Shift

The vector $\vec{a} = -\hat{i} + 2\hat{j} + \hat{k}$ is rotated through a right angle, passing through the y-axis in its way and the resulting vector is $\vec{b}$. Then the projection of $3\vec{a} + \sqrt{2}\vec{b}$ on $\vec{c} = 5\hat{i} + 4\hat{j} + 3\hat{k}$ is :

(A) $\sqrt{6}$ (B) $2\sqrt{3}$ (C) 1 (D) $3\sqrt{2}$

Open solution

Q92. JEE Main 2023 (Online) 24th January Evening Shift

Let $\vec{\alpha} = 4\hat{i} + 3\hat{j} + 5\hat{k}$ and $\vec{\beta} = \hat{i} + 2\hat{j} - 4\hat{k}$. Let $\vec{\beta_1}$ be parallel to $\vec{\alpha}$ and $\vec{\beta_2}$ be perpendicular to $\vec{\alpha}$. If $\vec{\beta} = \vec{\beta_1} + \vec{\beta_2}$, then the value of $5\vec{\beta_2} \cdot \left(\hat{i} + \hat{j} + \hat{k}\right)$ is :

(A) 9 (B) 7 (C) 6 (D) 11

Open solution

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Q93. JEE Main 2023 (Online) 24th January Morning Shift

Let PQR be a triangle. The points A, B and C are on the sides QR, RP and PQ respectively such that $\frac{QA}{AR} = \frac{RB}{BP} = \frac{PC}{CQ} = \frac{1}{2}$. Then $\frac{Area(\Delta PQR)}{Area(\Delta ABC)}$ is equal to :

Open solution

(A) $\frac{5}{2}$ (B) 4 (C) 2 (D) 3

Q94. JEE Main 2022 (Online) 29th July Evening Shift

Let $\vec{a}, \vec{b}, \vec{c}$ be three coplanar concurrent vectors such that the angle between any two of them is the same. If the product of their magnitudes is 14 and $(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) + (\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) + (\vec{c} \times \vec{a}) \cdot (\vec{a} \times \vec{b}) = 168$, then $|\vec{a}| + |\vec{b}| + |\vec{c}|$ is equal to :

Open solution

(A) 10 (B) 14 (C) 16 (D) 18

Q95. JEE Main 2022 (Online) 29th July Morning Shift

Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that the angle between them is $\frac{\pi}{4}$. If θ is the angle between the vectors $(\hat{a} + \hat{b})$ and $(\hat{a} + 2\hat{b} + 2(\hat{a} \times \hat{b}))$, then the value of $164 \cos^2 \theta$ is equal to :

Open solution

(A) $90 + 27\sqrt{2}$ (B) $45 + 18\sqrt{2}$ (C) $90 + 3\sqrt{2}$ (D) $54 + 90\sqrt{2}$

Q96. JEE Main 2022 (Online) 28th July Evening Shift

Let S be the set of all $a \in R$ for which the angle between the vectors $\vec{u} = a(\log_e b)\hat{i} - 6\hat{j} + 3\hat{k}$ and $\vec{v} = (\log_e b)\hat{i} + 2\hat{j} + 2a(\log_e b)\hat{k}$, $(b > 1)$ is acute. Then S is equal to :

Open solution

(A) $(-\infty, -\frac{4}{3})$ (B) Φ

(C) $(-\frac{4}{3}, 0)$ (D) $(\frac{12}{7}, \infty)$

Q97. JEE Main 2022 (Online) 28th July Morning Shift

Let the vectors $\vec{a} = (1+t)\hat{i} + (1-t)\hat{j} + \hat{k}, \vec{b} = (1-t)\hat{i} + (1+t)\hat{j} + 2\hat{k}$ and $\vec{c} = t\hat{i} - t\hat{j} + \hat{k}, t \in \mathbf{R}$ be such that for $\alpha, \beta, \gamma \in \mathbf{R}, \alpha\vec{a} + \beta\vec{b} + \gamma\vec{c} = \vec{0} \Rightarrow \alpha = \beta = \gamma = 0$. Then, the set of all values of t is :

Open solution

(A) a non-empty finite set (B) equal to $\mathbf{N}$

(C) equal to $\mathbf{R} - \{0\}$ (D) equal to $\mathbf{R}$

Q98. JEE Main 2022 (Online) 28th July Morning Shift

Let a vector $\vec{a}$ has magnitude 9. Let a vector $\vec{b}$ be such that for every $(x, y) \in \mathbf{R} \times \mathbf{R} - \{(0, 0)\}$, the vector $(x\vec{a} + y\vec{b})$ is perpendicular to the vector $(6y\vec{a} - 18x\vec{b})$. Then the value of $|\vec{a} \times \vec{b}|$ is equal to :

Open solution

(A) $9\sqrt{3}$ (B) $27\sqrt{3}$ (C) 9 (D) 81

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Q99. JEE Main 2022 (Online) 27th July Morning Shift

Let $\vec{a} = \alpha\hat{i} + \hat{j} + \beta\hat{k}$ and $\vec{b} = 3\hat{i} - 5\hat{j} + 4\hat{k}$ be two vectors, such that $\vec{a} \times \vec{b} = -\hat{i} + 9\hat{j} + 12\hat{k}$. Then the projection of $\vec{b} - 2\vec{a}$ on $\vec{b} + \vec{a}$ is equal to :

(A) 2 (B) $\frac{39}{5}$ (C) 9 (D) $\frac{46}{5}$

Open solution

Q100. JEE Main 2022 (Online) 26th July Morning Shift

Let $\overrightarrow{a} = \alpha\hat{i} + \hat{j} - \hat{k}$ and $\overrightarrow{b} = 2\hat{i} + \hat{j} - \alpha\hat{k}, \alpha > 0$. If the projection of $\overrightarrow{a} \times \overrightarrow{b}$ on the vector $-\hat{i} + 2\hat{j} - 2\hat{k}$ is 30, then α is equal to :

(A) $\frac{15}{2}$ (B) 8 (C) $\frac{13}{2}$ (D) 7

Open solution

Q101. JEE Main 2022 (Online) 25th July Evening Shift

Let $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$ and let $\vec{b}$ be a vector such that $\vec{a} \times \vec{b} = 2\hat{i} - \hat{k}$ and $\vec{a} \cdot \vec{b} = 3$. Then the projection of $\vec{b}$ on the vector $\vec{a} - \vec{b}$ is :

(A) $\frac{2}{\sqrt{21}}$ (B) $2\sqrt{\frac{3}{7}}$ (C) $\frac{2}{3}\sqrt{\frac{7}{3}}$ (D) $\frac{2}{3}$

Open solution

Q102. JEE Main 2022 (Online) 25th July Morning Shift

Let ABC be a triangle such that $\overrightarrow{BC} = \overrightarrow{a}, \overrightarrow{CA} = \overrightarrow{b}, \overrightarrow{AB} = \overrightarrow{c}, |\overrightarrow{a}| = 6\sqrt{2}, |\overrightarrow{b}| = 2\sqrt{3}$ and $\vec{b} \cdot \vec{c} = 12$. Consider the statements :

$(S1) : |(\overrightarrow{a} \times \overrightarrow{b}) + (\overrightarrow{c} \times \overrightarrow{b})| - |\vec{c}| = 6(2\sqrt{2} - 1)$

$(S2) : \angle ACB = \cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$

Then

(A) both (S1) and (S2) are true (B) only (S1) is true

(C) only (S2) is true (D) both (S1) and (S2) are false

Open solution

Q103. JEE Main 2022 (Online) 29th June Evening Shift

Let A, B, C be three points whose position vectors respectively are

$\overrightarrow{a} = \hat{i} + 4\hat{j} + 3\widehat{k}$

$\overrightarrow{b} = 2\hat{i} + \alpha\hat{j} + 4\widehat{k}, \alpha \in R$

$\overrightarrow{c} = 3\hat{i} - 2\hat{j} + 5\widehat{k}$

If α is the smallest positive integer for which $\overrightarrow{a}, \overrightarrow{b}, \overrightarrow{c}$ are noncollinear, then the length of the median, in ΔABC, through A is :

(A) $\frac{\sqrt{82}}{2}$ (B) $\frac{\sqrt{62}}{2}$ (C) $\frac{\sqrt{69}}{2}$ (D) $\frac{\sqrt{66}}{2}$

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Q104. JEE Main 2022 (Online) 29th June Morning Shift

Let $\vec{a}=\alpha\hat{i}+3\hat{j}-\hat{k}, \vec{b}=3\hat{i}-\beta\hat{j}+4\hat{k}$ and $\vec{c}=\hat{i}+2\hat{j}-2\hat{k}$ where $\alpha, \beta \in R$, be three vectors. If the projection of $\vec{a}$ on $\vec{c}$ is $\frac{10}{3}$ and $\vec{b}\times\vec{c}=-6\hat{i}+10\hat{j}+7\hat{k}$, then the value of $\alpha+\beta$ is equal to :

(A) 3 (B) 4 (C) 5 (D) 6

Open solution

Q105. JEE Main 2022 (Online) 28th June Evening Shift

Let $\vec{a}=\alpha\hat{i}+2\hat{j}-\hat{k}$ and $\vec{b}=-2\hat{i}+\alpha\hat{j}+\hat{k}$, where $\alpha \in R$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $\sqrt{15(\alpha^2+4)}$, then the value of $2\left|\vec{a}\right|^2+\left(\vec{a}.\vec{b}\right)\left|\vec{b}\right|^2$ is equal to :

(A) 10 (B) 7 (C) 9 (D) 14

Open solution

Q106. JEE Main 2022 (Online) 28th June Evening Shift

Let $\vec{a}$ be a vector which is perpendicular to the vector $3\hat{i}+\frac{1}{2}\hat{j}+2\hat{k}$. If $\vec{a}\times\left(2\hat{i}+\hat{k}\right)=2\hat{i}-13\hat{j}-4\hat{k}$, then the projection of the vector $\vec{a}$ on the vector $2\hat{i}+2\hat{j}+\hat{k}$ is :

(A) $\frac{1}{3}$ (B) 1 (C) $\frac{5}{3}$ (D) $\frac{7}{3}$

Open solution

Q107. JEE Main 2022 (Online) 27th June Evening Shift

Let $\vec{a}$ and $\vec{b}$ be the vectors along the diagonals of a parallelogram having area $2\sqrt{2}$. Let the angle between $\vec{a}$ and $\vec{b}$ be acute, $|\vec{a}|=1$, and $|\vec{a}.\vec{b}|=|\vec{a}\times\vec{b}|$. If $\vec{c}=2\sqrt{2}\left(\vec{a}\times\vec{b}\right)-2\vec{b}$, then an angle between $\vec{b}$ and $\vec{c}$ is :

(A) $\frac{\pi}{4}$ (B) $-\frac{\pi}{4}$ (C) $\frac{5\pi}{6}$ (D) $\frac{3\pi}{4}$

Open solution

Q108. JEE Main 2022 (Online) 27th June Morning Shift

Let $\vec{a}=\hat{i}+\hat{j}-\hat{k}$ and $\vec{c}=2\hat{i}-3\hat{j}+2\hat{k}$. Then the number of vectors $\vec{b}$ such that $\vec{b}\times\vec{c}=\vec{a}$ and $|\vec{b}| \in \{1, 2, ..., 10\}$ is :

(A) 0 (B) 1 (C) 2 (D) 3

Open solution

Q109. JEE Main 2022 (Online) 26th June Evening Shift

Let $\vec{a}=\hat{i}+\hat{j}+2\hat{k}, \vec{b}=2\hat{i}-3\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}+\hat{k}$ be three given vectors. Let $\vec{v}$ be a vector in the plane of $\vec{a}$ and $\vec{b}$ whose projection on $\vec{c}$ is $\frac{2}{\sqrt{3}}$. If $\vec{v}.\hat{j}=7$, then $\vec{v}.\left(\hat{i}+\hat{k}\right)$ is equal to :

(A) 6 (B) 7 (C) 8 (D) 9

Open solution

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Q110. JEE Main 2022 (Online) 25th June Morning Shift

Open solution

Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ $a_i > 0, i = 1, 2, 3$ be a vector which makes equal angles with the coordinate axes OX, OY and OZ. Also, let the projection of $\vec{a}$ on the vector $3\hat{i} + 4\hat{j}$ be 7. Let $\vec{b}$ be a vector obtained by rotating $\vec{a}$ with 90°. If $\vec{a}, \vec{b}$ and x-axis are coplanar, then projection of a vector $\vec{b}$ on $3\hat{i} + 4\hat{j}$ is equal to:

(A) $\sqrt{7}$ (B) $\sqrt{2}$ (C) 2 (D) 7

Q111. JEE Main 2022 (Online) 24th June Evening Shift

Open solution

Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that $|(\hat{a} + \hat{b}) + 2(\hat{a} \times \hat{b})| = 2$. If $\theta \in (0, \pi)$ is the angle between $\hat{a}$ and $\hat{b}$, then among the statements :

(S1) : $2|\hat{a} \times \hat{b}| = |\hat{a} - \hat{b}|$

(S2) : The projection of $\hat{a}$ on $(\hat{a} + \hat{b})$ is $\frac{1}{2}$

(A) Only (S1) is true.
(B) Only (S2) is true.
(C) Both (S1) and (S2) are true.
(D) Both (S1) and (S2) are false.

Q112. JEE Main 2022 (Online) 24th June Morning Shift

Open solution

Let $\hat{a}, \hat{b}$ be unit vectors. If $\vec{c}$ be a vector such that the angle between $\hat{a}$ and $\vec{c}$ is $\frac{\pi}{12}$, and $\hat{b} = \vec{c} + 2\left(\vec{c} \times \hat{a}\right)$, then $\left|6\vec{c}\right|^2$ is equal to :

(A) $6\left(3 - \sqrt{3}\right)$ (B) $3 + \sqrt{3}$ (C) $6\left(3 + \sqrt{3}\right)$ (D) $6\left(\sqrt{3} + 1\right)$

Q113. JEE Main 2021 (Online) 31st August Morning Shift

Open solution

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $\left|2\vec{a} + 3\vec{b}\right| = \left|3\vec{a} + \vec{b}\right|$ and the angle between $\vec{a}$ and $\vec{b}$ is 60°. If $\frac{1}{8}\vec{a}$ is a unit vector, then $\left|\vec{b}\right|$ is equal to :

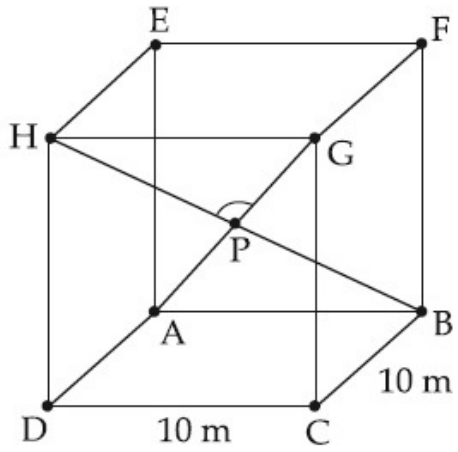
(A) 4 (B) 6 (C) 5 (D) 8

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Q114. JEE Main 2021 (Online) 26th August Evening Shift

A hall has a square floor of dimension 10 m $\times$ 10 m (see the figure) and vertical walls. If the angle GPH between the diagonals AG and BH is $\cos^{-1}\frac{1}{5}$, then the height of the hall (in meters) is :

Open solution



(A) 5 (B) $2\sqrt{10}$ (C) $5\sqrt{3}$ (D) $5\sqrt{2}$

Q115. JEE Main 2021 (Online) 27th July Morning Shift

Let $\vec{a}=\hat{i}+\hat{j}+2\hat{k}$ and $\vec{b}=-\hat{i}+2\hat{j}+3\hat{k}$. Then the vector product $\left(\vec{a}+\vec{b}\right)\times\left(\left(\vec{a}\times\left(\left(\vec{a}-\vec{b}\right)\times\vec{b}\right)\right)\times\vec{b}\right)$ is equal to :

Open solution

(A) $5(34\hat{i}-5\hat{j}+3\hat{k})$ (B) $7(34\hat{i}-5\hat{j}+3\hat{k})$

(C) $7(30\hat{i}-5\hat{j}+7\hat{k})$ (D) $5(30\hat{i}-5\hat{j}+7\hat{k})$

Q116. JEE Main 2021 (Online) 25th July Evening Shift

Let a, b and c be distinct positive numbers. If the vectors $a\hat{i}+a\hat{j}+c\hat{k},\hat{i}+\hat{k}$ and $c\hat{i}+c\hat{j}+b\hat{k}$ are co-planar, then c is equal to :

Open solution

(A) $\frac{2}{\frac{1}{a}+\frac{1}{b}}$ (B) $\frac{a+b}{2}$

(C) $\frac{1}{a}+\frac{1}{b}$ (D) $\sqrt{ab}$

Q117. JEE Main 2021 (Online) 25th July Evening Shift

If $\left|\vec{a}\right|=2,\left|\vec{b}\right|=5$ and $\left|\vec{a}\times\vec{b}\right|$ = 8, then $\left|\vec{a}.\vec{b}\right|$ is equal to :

Open solution

(A) 6 (B) 4 (C) 3 (D) 5

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Q118. JEE Main 2021 (Online) 20th July Evening Shift

In a triangle ABC, if $\left|\overrightarrow{BC}\right| = 3$, $\left|\overrightarrow{CA}\right| = 5$ and $\left|\overrightarrow{BA}\right| = 7$, then the projection of the vector $\overrightarrow{BA}$ on $\overrightarrow{BC}$ is equal to :

Open solution

(A) $\frac{19}{2}$ (B) $\frac{13}{2}$ (C) $\frac{11}{2}$ (D) $\frac{15}{2}$

Q119. JEE Main 2021 (Online) 18th March Evening Shift

Let $\vec{a}$ and $\vec{b}$ be two non-zero vectors perpendicular to each other and $|\vec{a}| = |\vec{b}|$. If $|\vec{a} \times \vec{b}| = |\vec{a}|$, then the angle between the vectors $\left(\vec{a} + \vec{b} + \left(\vec{a} \times \vec{b}\right)\right)$ and $\vec{a}$ is equal to :

Open solution

(A) $\sin^{-1}\left(\frac{1}{\sqrt{6}}\right)$ (B) $\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(C) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (D) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Q120. JEE Main 2021 (Online) 18th March Evening Shift

In a triangle ABC, if $|\overrightarrow{BC}| = 8, |\overrightarrow{CA}| = 7, |\overrightarrow{AB}| = 10$, then the projection of the vector $\overrightarrow{AB}$ on $\overrightarrow{AC}$ is equal to :

Open solution

(A) $\frac{25}{4}$ (B) $\frac{127}{20}$ (C) $\frac{85}{14}$ (D) $\frac{115}{16}$

Q121. JEE Main 2021 (Online) 18th March Morning Shift

A vector $\vec{a}$ has components 3p and 1 with respect to a rectangular cartesian system. This system is rotated through a certain angle about the origin in the counter clockwise sense. If, with respect to new system, $\vec{a}$ has components p + 1 and $\sqrt{10}$, then the value of p is equal to :

Open solution

(A) 1 (B) $-\frac{5}{4}$ (C) $\frac{4}{5}$ (D) -1

Q122. JEE Main 2021 (Online) 17th March Evening Shift

Let O be the origin. Let $\overrightarrow{OP} = x\hat{i} + y\hat{j} - \hat{k}$ and $\overrightarrow{OQ} = -\hat{i} + 2\hat{j} + 3x\hat{k}$, x, y∈R, x > 0, be such that $\left|\overrightarrow{PQ}\right| = \sqrt{20}$ and the vector $\overrightarrow{OP}$ is perpendicular $\overrightarrow{OQ}$. If $\overrightarrow{OR} = 3\hat{i} + z\hat{j} - 7\hat{k}$, z∈R, is coplanar with $\overrightarrow{OP}$ and $\overrightarrow{OQ}$, then the value of $x^2 + y^2 + z^2$ is equal to :

Open solution

(A) 2 (B) 9 (C) 7 (D) 1

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Q123. JEE Main 2021 (Online) 16th March Evening Shift

Let $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 2\hat{i} - 3\hat{j} + 5\hat{k}$. If $\vec{r} \times \vec{a} = \vec{b} \times \vec{r}$, $\vec{r} \cdot \left(\alpha\hat{i} + 2\hat{j} + \hat{k}\right) = 3$ and $\vec{r} \cdot \left(2\hat{i} + 5\hat{j} - \alpha\hat{k}\right) = -1$, $\alpha \in R$, then the value of $\alpha + \left|\vec{r}\right|^2$ is equal to :

Open solution

(A) 13 (B) 11 (C) 9 (D) 15

Q124. JEE Main 2021 (Online) 16th March Morning Shift

Let a vector $\alpha\hat{i} + \beta\hat{j}$ be obtained by rotating the vector $\sqrt{3}\hat{i} + \hat{j}$ by an angle 45° about the origin in counterclockwise direction in the first quadrant. Then the area of triangle having vertices (α, β), (0, β) and (0, 0) is equal to :

Open solution

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) 1 (D) $2\sqrt{2}$

Q125. JEE Main 2021 (Online) 26th February Evening Shift

If vectors $\vec{a_1} = x\hat{i} - \hat{j} + \hat{k}$ and $\vec{a_2} = \hat{i} + y\hat{j} + z\hat{k}$ are collinear, then a possible unit vector parallel to the vector $x\hat{i} + y\hat{j} + z\hat{k}$ is :

Open solution

(A) $\frac{1}{\sqrt{3}}\left(\hat{i} - \hat{j} + \hat{k}\right)$ (B) $\frac{1}{\sqrt{2}}\left(-\hat{j} + \hat{k}\right)$
(C) $\frac{1}{\sqrt{2}}\left(\hat{i} - \hat{j}\right)$ (D) $\frac{1}{\sqrt{3}}\left(\hat{i} + \hat{j} - \hat{k}\right)$

Q126. JEE Main 2020 (Online) 3rd September Evening Slot

Let a, b, c $\in$ R be such that $a^2 + b^2 + c^2 = 1$. If
$a\cos\theta = b\cos\left(\theta + \frac{2\pi}{3}\right) = c\cos\left(\theta + \frac{4\pi}{3}\right)$,
where $\theta = \frac{\pi}{9}$, then the angle between the vectors $a\hat{i} + b\hat{j} + c\hat{k}$ and $b\hat{i} + c\hat{j} + a\hat{k}$ is :

Open solution

(A) 0 (B) $\frac{\pi}{9}$ (C) $\frac{2\pi}{3}$ (D) $\frac{\pi}{2}$

Q127. JEE Main 2020 (Online) 8th January Evening Slot

Let $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ be two vectors. If $\vec{c}$ is a vector such that $\vec{b} \times \vec{c} = \vec{b} \times \vec{a}$ and $\vec{c} \cdot \vec{a} = 0$, then $\vec{c} \cdot \vec{b}$ is equal to

Open solution

(A) $-\frac{1}{2}$ (B) $-\frac{3}{2}$ (C) $\frac{1}{2}$ (D) -1

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Q128. JEE Main 2020 (Online) 7th January Evening Slot

Open solution

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three unit vectors such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$. If $\lambda=\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}$ and $\vec{d}=\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}$, then the ordered pair, $\left(\lambda,\vec{d}\right)$ is equal to :

(A) $\left(\frac{3}{2},3\vec{a}\times\vec{c}\right)$

(B) $\left(-\frac{3}{2},3\vec{c}\times\vec{b}\right)$

(C) $\left(-\frac{3}{2},3\vec{a}\times\vec{b}\right)$

(D) $\left(\frac{3}{2},3\vec{b}\times\vec{c}\right)$

Q129. JEE Main 2020 (Online) 7th January Morning Slot

Open solution

A vector $\vec{a}=\alpha\hat{i}+2\hat{j}+\beta\hat{k}\,(\alpha,\beta\in R)$ lies in the plane of the vectors, $\vec{b}=\hat{i}+\hat{j}$ and $\vec{c}=\hat{i}-\hat{j}+4\hat{k}$. If $\vec{a}$ bisects the angle between $\vec{b}$ and $\vec{c}$, then:

(A) $\vec{a}.\hat{i}+3=0$

(B) $\vec{a}.\hat{k}-4=0$

(C) $\vec{a}.\hat{i}+1=0$

(D) $\vec{a}.\hat{k}+2=0$

Q130. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

Let $\vec{a}=3\hat{i}+2\hat{j}+2\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}-2\hat{k}$ be two vectors. If a vector perpendicular to both the vectors $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$ has the magnitude 12 then one such vector is :

(A) $4\left(2\hat{i}-2\hat{j}-\hat{k}\right)$

(B) $4\left(-2\hat{i}-2\hat{j}+\hat{k}\right)$

(C) $4\left(2\hat{i}+2\hat{j}+\hat{k}\right)$

(D) $4\left(2\hat{i}+2\hat{j}-\hat{k}\right)$

Q131. JEE Main 2019 (Online) 10th April Morning Slot

Open solution

Let A (3, 0, −1), B(2, 10, 6) and C(1, 2, 1) be the vertices of a triangle and M be the midpoint of AC. If G divides BM in the ratio, 2 : 1, then cos (∠GOA) (O being the origin) is equal to :

(A) $\frac{1}{\sqrt{15}}$

(B) $\frac{1}{6\sqrt{10}}$

(C) $\frac{1}{\sqrt{30}}$

(D) $\frac{1}{2\sqrt{15}}$

Q132. JEE Main 2019 (Online) 9th April Evening Slot

Open solution

If a unit vector $\vec{a}$ makes angles $\pi/3$ with $\hat{i}$, $\pi/4$ with $\hat{j}$ and $\theta\in(0,\pi)$ with $\hat{k}$, then a value of θ is :-

(A) $\frac{5\pi}{6}$

(B) $\frac{5\pi}{12}$

(C) $\frac{2\pi}{3}$

(D) $\frac{\pi}{4}$

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Q133. JEE Main 2019 (Online) 9th April Morning Slot

Let $\vec{\alpha} = 3\hat{i} + \hat{j}$ and $\vec{\beta} = 2\hat{i} - \hat{j} + 3\hat{k}$. If $\vec{\beta} = \vec{\beta_1} - \vec{\beta_2}$, where $\vec{\beta_1}$ is parallel to $\vec{\alpha}$ and $\vec{\beta_2}$ is perpendicular to $\vec{\alpha}$, then $\vec{\beta_1} \times \vec{\beta_2}$ is equal to

(A) $3\hat{i} - 9\hat{j} - 5\hat{k}$

(B) $\frac{1}{2}(-3\hat{i} + 9\hat{j} + 5\hat{k})$

(C) $-3\hat{i} + 9\hat{j} + 5\hat{k}$

(D) $\frac{1}{2}(3\hat{i} - 9\hat{j} + 5\hat{k})$

Open solution

Q134. JEE Main 2019 (Online) 8th April Evening Slot

Let $\vec{a} = 3\hat{i} + 2\hat{j} + x\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, for some real x. Then $|\vec{a} \times \vec{b}|$ = r is possible if :

(A) $0 < r < \sqrt{\frac{3}{2}}$

(B) $3\sqrt{\frac{3}{2}} < r < 5\sqrt{\frac{3}{2}}$

(C) $r \geq 5\sqrt{\frac{3}{2}}$

(D) $\sqrt{\frac{3}{2}} < r \leq 3\sqrt{\frac{3}{2}}$

Open solution

Q135. JEE Main 2019 (Online) 11th January Evening Slot

Let $\sqrt{3}\hat{i} + \hat{j}$, $\hat{i} + \sqrt{3}\hat{j}$ and $\beta\hat{i} + (1-\beta)\hat{j}$ respectively be the position vectors of the points A, B and C with respect to the origin O. If the distance of C from the bisector of the acute angle between OA and OB is $\frac{3}{\sqrt{2}}$, then the sum of all possible values of β is :

(A) 4

(B) 1

(C) 2

(D) 3

Open solution

Q136. JEE Main 2019 (Online) 11th January Morning Slot

Let $\vec{a} = \hat{i} + 2\hat{j} + 4\hat{k}$, $\vec{b} = \hat{i} + \lambda\hat{j} + 4\hat{k}$ and $\vec{c} = 2\hat{i} + 4\hat{j} + (\lambda^2 - 1)\hat{k}$ be coplanar vectors. Then the non-zero vector $\vec{a} \times \vec{c}$ is :

(A) $-10\hat{i} - 5\hat{j}$

(B) $-10\hat{i} + 5\hat{j}$

(C) $-14\hat{i} + 5\hat{j}$

(D) $-14\hat{i} - 5\hat{j}$

Open solution

Q137. JEE Main 2019 (Online) 10th January Evening Slot

If $\vec{\alpha} = (\lambda - 2)\vec{a} + \vec{b}$ and $\vec{\beta} = (4\lambda - 2)\vec{a} + 3\vec{b}$ be two given vectors $\vec{a}$ and $\vec{b}$ are non-collinear. The value of λ for which vectors $\vec{\alpha}$ and $\vec{\beta}$ are collinear, is -

(A) 4

(B) 3

(C) -3

(D) -4

Open solution

Q138. JEE Main 2019 (Online) 10th January Morning Slot

Let $\vec{a} = 2\hat{i} + \lambda_1\hat{j} + 3\hat{k}$, $\vec{b} = 4\hat{i} + (3 - \lambda_2)\hat{j} + 6\hat{k}$, and $\vec{c} = 3\hat{i} + 6\hat{j} + (\lambda_3 - 1)\hat{k}$ be three vectors such that $\vec{b} = 2\vec{a}$ and $\vec{a}$ is perpendicular to $\vec{c}$. Then a possible value of $(\lambda_1, \lambda_2, \lambda_3)$ is :

(A) (1, 5, 1)

(B) (1, 3, 1)

(C) $\left(-\frac{1}{2}, 4, 0\right)$

(D) $\left(\frac{1}{2}, 4, -2\right)$

Open solution

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Q139. JEE Main 2019 (Online) 9th January Evening Slot

Let $\vec{a}=\hat{i}+\hat{j}+\sqrt{2}\hat{k}$, $\vec{b}=b_1\hat{i}+b_2\hat{j}+\sqrt{2}\hat{k}$, $\vec{c}=5\hat{i}+\hat{j}+\sqrt{2}\hat{k}$ be three vectors such that the projection vector of $\vec{b}$ on $\vec{a}$ is $\vec{a}$.
If $\vec{a}+\vec{b}$ is perpendicular to $\vec{c}$, then $|\vec{b}|$ is equal to :

(A) $\sqrt{32}$ (B) 6 (C) $\sqrt{22}$ (D) 4

Open solution

Q140. JEE Main 2018 (Online) 16th April Morning Slot

Let $\vec{a}=\hat{i}+\hat{j}+\hat{k}, \vec{c}=\hat{j}-\hat{k}$ and a vector $\vec{b}$ be such that $\vec{a}\times\vec{b}=\vec{c}$ and $\vec{a}.\vec{b}=3$. Then $|\vec{b}|$ equals :

(A) $\frac{11}{3}$ (B) $\frac{11}{\sqrt{3}}$ (C) $\sqrt{\frac{11}{3}}$ (D) $\frac{\sqrt{11}}{3}$

Open solution

Q141. JEE Main 2018 (Offline)

Let $\vec{u}$ be a vector coplanar with the vectors $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$ and $\vec{b}=\hat{j}+\hat{k}$. If $\vec{u}$ is perpendicular to $\vec{a}$ and $\vec{u}.\vec{b}=24$, then $|\vec{u}|^2$ is equal to

(A) 336 (B) 315 (C) 256 (D) 84

Open solution

Q142. JEE Main 2018 (Online) 15th April Evening Slot

If the position vectors of the vertices A, B and C of a Δ ABC are respectively $4\hat{i}+7\hat{j}+8\hat{k}$, $2\hat{i}+3\hat{j}+4\hat{k}$, and $2\hat{i}+5\hat{j}+7\hat{k}$, then the position vectors of the point, where the bisector of $\angle$A meets BC is :

(A) $\frac{1}{2}\left(4\hat{i}+8\hat{j}+11\hat{k}\right)$ (B) $\frac{1}{3}\left(6\hat{i}+11\hat{j}+15\hat{k}\right)$
(C) $\frac{1}{3}\left(6\hat{i}+13\hat{j}+18\hat{k}\right)$ (D) $\frac{1}{4}\left(8\hat{i}+14\hat{j}+19\hat{k}\right)$

Open solution

Q143. JEE Main 2018 (Online) 15th April Morning Slot

If $\vec{a}$, $\vec{b}$, and $\vec{C}$ are unit vectors such that $\vec{a}+2\vec{b}+2\vec{c}=\vec{0}$, then $|\vec{a}\times\vec{c}|$ is equal to :

(A) $\frac{\sqrt{15}}{4}$ (B) $\frac{1}{4}$ (C) $\frac{15}{16}$ (D) $\frac{\sqrt{15}}{16}$

Open solution

Q144. JEE Main 2017 (Online) 9th April Morning Slot

If the vector $\vec{b}=3\hat{j}+4\hat{k}$ is written as the sum of a vector $\vec{b_1}$, parallel to $\vec{a}=\hat{i}+\hat{j}$ and a vector $\vec{b_2}$, perpendicular to $\vec{a}$, then $\vec{b_1}\times\vec{b_2}$ is equal to :

(A) $-3\hat{i}+3\hat{j}-9\hat{k}$ (B) $6\hat{i}-6\hat{j}+\frac{9}{2}\hat{k}$
(C) $-6\hat{i}+6\hat{j}-\frac{9}{2}\hat{k}$ (D) $3\hat{i}-3\hat{j}+9\hat{k}$

Open solution

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Q145. JEE Main 2017 (Online) 8th April Morning Slot

The area (in sq. units) of the parallelogram whose diagonals are along the vectors $8\hat{i} - 6\hat{j}$ and $3\hat{i} + 4\hat{j} - 12\hat{k}$, is :

(A) 26 (B) 65 (C) 20 (D) 52

Open solution

Q146. JEE Main 2017 (Offline)

Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$.

Let $\vec{c}$ be a vector such that $|\vec{c} - \vec{a}| = 3$,

$\left|\left(\vec{a} \times \vec{b}\right) \times \vec{c}\right| = 3$ and the angle between $\vec{c}$ and $\vec{a} \times \vec{b}$ is 30°.

Then $\vec{a}.\vec{c}$ is equal to :

(A) 2 (B) 5 (C) $\frac{1}{8}$ (D) $\frac{25}{8}$

Open solution

Q147. JEE Main 2016 (Online) 10th April Morning Slot

Let ABC be a triangle whose circumcentre is at P. If the position vectors of A, B, C and P are $\vec{a}, \vec{b}, \vec{c}$ and $\frac{\vec{a}+\vec{b}+\vec{c}}{4}$ respectively, then the position vector of the orthocentre of this triangle, is :

(A) $\vec{a} + \vec{b} + \vec{c}$

(B) $-\left(\frac{\vec{a}+\vec{b}+\vec{c}}{2}\right)$

(C) $\vec{0}$

(D) $\left(\frac{\vec{a}+\vec{b}+\vec{c}}{2}\right)$

Open solution

Q148. JEE Main 2016 (Online) 9th April Morning Slot

In a triangle ABC, right angled at the vertex A, if the position vectors of A, B and C are respectively $3\hat{i}$ + $\hat{j} - \hat{k}$, $-\hat{i}$ + $3\hat{j}$ + p$\hat{k}$ and $5\hat{i}$ + q$\hat{j} - 4\hat{k}$, then the point (p, q) lies on a line :

(A) parallel to x-axis.

(B) parallel to y-axis.

(C) making an acute angle with the positive direction of x-axis.

(D) making an obtuse angle with the positive direction of x-axis.

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Q149. JEE Main 2015 (Online) 11th April Morning Slot

In a parallelogram ABCD,

$\left|\overrightarrow{AB}\right|$ = a, $\left|\overrightarrow{AD}\right|$ = b and $\left|\overrightarrow{AC}\right|$ = c,

then $\overrightarrow{DA}.\overrightarrow{AB}$ has the value :

(A) $\frac{1}{2}$ ($a^2 - b^2 + c^2$)

(B) $\frac{1}{2}$ ($a^2 + b^2 - c^2$)

(C) $\frac{1}{3}$ ($b^2 + c^2 - a^2$)

(D) $\frac{1}{2}$ ($a^2 + b^2 + c^2$)

Open solution

Q150. JEE Main 2015 (Online) 10th April Morning Slot

Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that $\left|\vec{a}+\vec{b}\right| = \sqrt{3}$.

If $\vec{c} = \vec{a} + 2\vec{b} + 3(\vec{a} \times \vec{b})$,

then $2\left|\vec{c}\right|$ is equal to :

(A) $\sqrt{55}$

(B) $\sqrt{51}$

(C) $\sqrt{43}$

(D) $\sqrt{37}$

Open solution

Q151. JEE Main 2014 (Online) 19th April Morning Slot

If $\vec{x} = 3\hat{i} - 6\hat{j} - \hat{k}$, $\vec{y} = \hat{i} + 4\hat{j} - 3\hat{k}$, and $\vec{z} = 3\hat{i} - 4\hat{j} - 12\hat{k}$, then the magnitude of the projection of $\vec{x} \times \vec{y}$ on $\vec{z}$ is :

(A) 12

(B) 15

(C) 14

(D) 13

Open solution

Q152. JEE Main 2014 (Online) 12th April Morning Slot

If $\hat{x}$, $\hat{y}$ and $\hat{z}$ are three unit vectors in three-dimensional space, then the minimum value of

$|\hat{x}+\hat{y}|^2 + |\hat{y}+\hat{z}|^2 + |\hat{z}+\hat{x}|^2$ is :

(A) $\frac{3}{2}$

(B) 3

(C) $3\sqrt{3}$

(D) 6

Open solution

Q153. JEE Main 2014 (Online) 11th April Morning Slot

If $\left|\vec{c}\right|^2$ = 60 and $\vec{c} \times (\hat{i} + 2\hat{j} + 5\hat{k}) = \vec{0}$,

then a value of $\vec{c}.(-7\hat{i} + 2\hat{j} + 3\hat{k})$ is :

(A) $4\sqrt{2}$

(B) 12

(C) 24

(D) $12\sqrt{2}$

Open solution

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Q154. JEE Main 2014 (Online) 9th April Morning Slot

If $\left|\vec{a}\right| = 2$, $\left|\vec{b}\right| = 3$ and $\left|2\vec{a} - \vec{b}\right| = 5$, then $\left|2\vec{a} + \vec{b}\right|$ equals :

(A) 17 (B) 7 (C) 5 (D) 1

Open solution

Q155. JEE Main 2013 (Online) 25th April Morning Slot

Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}, \vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that

$\vec{a} \bullet \vec{c} = \left|\vec{c}\right|, \left|\vec{c} - \vec{a}\right| = 2\sqrt{2}$ and the angle between $\vec{a} \times \vec{b}$ and $\vec{c}$ is 30°, then $\left|\left(\vec{a} \times \vec{b}\right) \times \vec{c}\right|$ equals:

(A) $\frac{1}{2}$ (B) $\frac{3\sqrt{3}}{2}$ (C) 3 (D) $\frac{3}{2}$

Open solution

Q156. JEE Main 2013 (Online) 23th April Morning Slot

If $\vec{a}$ and $\vec{b}$ are non-collinear vectors, then the value of α for which the vectors $\vec{u} = (\alpha - 2)\vec{a} + \vec{b}$ and $\vec{v} = (2 + 3\alpha)\vec{a} - 3\vec{b}$ are collinear is :

(A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) $-\frac{3}{2}$ (D) $-\frac{2}{3}$

Open solution

Q157. JEE Main 2013 (Online) 22th April Morning Slot

If $\hat{a}, \hat{b}$ and $\hat{c}$ are unit vectors satisfying $\hat{a} - \sqrt{3}\hat{b} + \hat{c} = \vec{0}$, then the angle between the vectors $\hat{a}$ and $\hat{c}$ is:

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{2}$

Open solution

Q158. JEE Main 2013 (Online) 9th April Morning Slot

Let $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} - 2\hat{k}$ be three vectors. A vector of the type $\vec{b} + \lambda\vec{c}$ for some scalar λ, whose projection on $\vec{a}$ is of magnitude $\sqrt{\frac{2}{3}}$ is :

(A) $2\hat{i} + \hat{j} + 5\hat{k}$ (B) $2\hat{i} + 3\hat{j} - 3\hat{k}$

(C) $2\hat{i} - \hat{j} + 5\hat{k}$ (D) $2\hat{i} + 3\hat{j} + 3\hat{k}$

Open solution

Q159. JEE Main 2013 (Offline)

If the vectors $\overrightarrow{AB} = 3\hat{i} + 4\hat{k}$ and $\overrightarrow{AC} = 5\hat{i} - 2\hat{j} + 4\hat{k}$ are the sides of a triangle ABC, then the length of the median through A is :

(A) $\sqrt{18}$ (B) $\sqrt{72}$ (C) $\sqrt{33}$ (D) $\sqrt{45}$

Open solution

Q160. AIEEE 2012 (Online) 19th May

If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, $|\vec{a}| = 3$, $|\vec{b}| = 5$ and $|\vec{c}| = 7$, then the angle between $\vec{a}$ and $\vec{b}$ is

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{2}$

Open solution

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Q161. AIEEE 2012 (Online) 7th May

$ABCD$ is parallelogram. The position vectors of A and C are respectively, $3\hat{i}+3\hat{j}+5\hat{k}$ and $\hat{i}-5\hat{j}-5\hat{k}$. If M is the midpoint of the diagonal DB, then the magnitude of the projection of $\overrightarrow{OM}$ on $\overrightarrow{OC}$, where O is the origin, is

Open solution

(A) $7\sqrt{51}$ (B) $\frac{7}{\sqrt{50}}$ (C) $7\sqrt{50}$ (D) $\frac{7}{\sqrt{51}}$

Q162. AIEEE 2012

Let $\vec{a}$ and $\vec{b}$ be two unit vectors. If the vectors $\vec{c}=\hat{a}+2\hat{b}$ and $\vec{d}=5\hat{a}-4\hat{b}$ are perpendicular to each other, then the angle between $\vec{a}$ and $\vec{b}$ is :

Open solution

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

Q163. AIEEE 2012

Let $ABCD$ be a parallelogram such that $\overrightarrow{AB}=\vec{q}, \overrightarrow{AD}=\vec{p}$ and $\angle BAD$ be an acute angle. If $\vec{r}$ is the vector that coincide with the altitude directed from the vertex B to the side AD, then $\vec{r}$ is given by :

Open solution

(A) $\vec{r}=3\vec{q}-\frac{3(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$

(B) $\vec{r}=-\vec{q}+\frac{(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$

(C) $\vec{r}=\vec{q}-\frac{(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$

(D) $\vec{r}=-3\vec{q}+\frac{3(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$

Q164. AIEEE 2011

The vectors $\vec{a}$ and $\vec{b}$ are not perpendicular and $\vec{c}$ and $\vec{d}$ are two vectors satisfying $\vec{b}\times\vec{c}=\vec{b}\times\vec{d}$ and $\vec{a}.\vec{d}=0$. Then the vector $\vec{d}$ is equal to :

Open solution

(A) $\vec{c}+\left(\frac{\vec{a}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{b}$

(B) $\vec{b}+\left(\frac{\vec{b}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{c}$

(C) $\vec{c}-\left(\frac{\vec{a}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{b}$

(D) $\vec{b}-\left(\frac{\vec{b}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{c}$

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Q165. AIEEE 2011

Let $\vec{a}, \vec{b}, \vec{c}$ be three non-zero vectors which are pairwise non-collinear. If $\vec{a} + 3\vec{b}$ is collinear with $\vec{c}$ and $\vec{b} + 2\vec{c}$ is collinear with $\vec{a}$, then $\vec{a} + 3\vec{b} + 6\vec{c}$ is :

(A) $\vec{a} + \vec{c}$

(B) $\vec{c}$

(C) $\vec{a}$

(D) $\vec{0}$

Open solution

Q166. AIEEE 2010

If the vectors $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} + \hat{k}$ and $\vec{c} = \lambda\hat{i} + \hat{j} + \mu\hat{k}$ are mutually orthogonal, then (λ, μ) is equal to :

(A) $(2, -3)$ (B) $(-2, 3)$ (C) $(3, -2)$ (D) $(-3, 2)$

Open solution

Q167. AIEEE 2008

The non-zero vectors are $\vec{a}, \vec{b}$ and $\vec{c}$ are related by $\vec{a} = 8\vec{b}$ and $\vec{c} = -7\vec{b}$. Then the angle between $\vec{a}$ and $\vec{c}$ is :

(A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) π

Open solution

Q168. AIEEE 2007

If $\hat{u}$ and $\hat{v}$ are unit vectors and θ is the acute angle between them, then $2\hat{u} \times 3\hat{v}$ is a unit vector for :

(A) no value of θ (B) exactly one value of θ

(C) exactly two values of θ (D) more than two values of θ

Open solution

Q169. AIEEE 2007

A body weighing 13 kg is suspended by two strings 5 m and 12 m long, their other ends being fastened to the extremities of a rod 13 m long. If the rod be so held that the body hangs immediately below the middle point. The tensions in the strings are

(A) 12 kg and 13 kg (B) 5 kg and 5 kg (C) 5 kg and 12 kg (D) 5 kg and 13 kg

Open solution

Q170. AIEEE 2007

The resultant of two forces P N and 3 N is a force of 7 N. If the direction of 3 N force were reversed, the resultant would be $\sqrt{19}$ N. The value of P is

(A) 5 N (B) 6 N (C) 3 N (D) 4 N

Open solution

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Q171. AIEEE 2006

The values of a, for which the points A, B, C with position vectors $2\hat{i}-\hat{j}+\hat{k}$, $\hat{i}-3\hat{j}-5\hat{k}$ and $a\hat{i}-3\hat{j}+\hat{k}$ respectively are the vertices of a right angled triangle with $C=\frac{\pi}{2}$ are :

(A) 2 and 1 (B) -2 and -1 (C) -2 and 1 (D) 2 and -1

Open solution

Q172. AIEEE 2006

ABC is a triangle, right angled at A. The resultant of the forces acting along $\overrightarrow{AB}$, $\overrightarrow{AC}$ with magnitudes $\frac{1}{AB}$ and $\frac{1}{AC}$ respectively is the force along $\overrightarrow{AD}$, where D is the foot of the perpendicular from A onto BC. The magnitude of the resultant is

(A) $\frac{AB^2+AC^2}{(AB)^2(AC)^2}$ (B) $\frac{(AB)(AC)}{AB+AC}$ (C) $\frac{1}{AB}+\frac{1}{AC}$ (D) $\frac{1}{AD}$

Open solution

Q173. AIEEE 2006

A particle has two velocities of equal magnitude inclined to each other at an angle θ. If one of them is halved, the angle between the other and the original resultant velocity is bisected by the new resultant. Then θ is

(A) 90° (B) 120° (C) 45° (D) 60°

Open solution

Q174. AIEEE 2005

If C is the mid point of AB and P is any point outside AB, then :

(A) $\overrightarrow{PA}+\overrightarrow{PB}=2\overrightarrow{PC}$

(B) $\overrightarrow{PA}+\overrightarrow{PB}=\overrightarrow{PC}$

(C) $\overrightarrow{PA}+\overrightarrow{PB}=2\overrightarrow{PC}=\vec{0}$

(D) $\overrightarrow{PA}+\overrightarrow{PB}=\overrightarrow{PC}=\vec{0}$

Open solution

Q175. AIEEE 2005

Let a, b and c be distinct non-negative numbers. If the vectors $a\hat{i}+a\hat{j}+c\hat{k}$, $\hat{i}+\hat{k}$ and $c\hat{i}+c\hat{j}+b\hat{k}$ lie in a plane, then c is :

(A) the Geometric Mean of a and b (B) the Arithmetic Mean of a and b

(C) equal to zero (D) the Harmonic Mean of a and b

Open solution

Q176. AIEEE 2005

For any vector $\vec{a}$, the value of $\left(\vec{a}\times\hat{i}\right)^2+\left(\vec{a}\times\hat{j}\right)^2+\left(\vec{a}\times\hat{k}\right)^2$ is equal to :

(A) $3\vec{a}^2$ (B) $\vec{a}^2$ (C) $2\vec{a}^2$ (D) $4\vec{a}^2$

Open solution

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Q177. AIEEE 2005

A and B are two like parallel forces. A couple of moment H lies in the plane of A and B and is combined with them. The resultant of A and B after combining is displaced through a distance

(A) $\frac{2H}{A-B}$ (B) $\frac{H}{A+B}$ (C) $\frac{H}{2(A+B)}$ (D) $\frac{H}{A-B}$

Open solution

Q178. AIEEE 2005

The resultant R of two forces acting on a particle is at right angles to one of them and its magnitude is one third of the other force. The ratio of larger force to smaller one is

(A) $2:1$ (B) $3:\sqrt{2}$ (C) $3:2$ (D) $3:2\sqrt{2}$

Open solution

Q179. AIEEE 2004

A particle acted on by constant forces $4\hat{i}+\hat{j}-3\hat{k}$ and $3\hat{i}+\hat{j}-\hat{k}$ is displaced from the point $\hat{i}+2\hat{j}+3\hat{k}$ to the point $5\hat{i}+4\hat{j}+\hat{k}$. The total work done by the forces is :

(A) 50 units (B) 20 units (C) 30 units (D) 40 units

Open solution

Q180. AIEEE 2004

Let $\vec{a},\vec{b}$ and $\vec{c}$ be three non-zero vectors such that no two of these are collinear. If the vector $\vec{a}+2\vec{b}$ is collinear with $\vec{c}$ and $\vec{b}+3\vec{c}$ is collinear with $\vec{a}$ (λ being some non-zero scalar) then $\vec{a}+2\vec{b}+6\vec{c}$ equals to :

(A) $\vec{0}$ (B) $\lambda\vec{b}$ (C) $\lambda\vec{c}$ (D) $\lambda\vec{a}$

Open solution

Q181. AIEEE 2004

Let $\vec{u},\vec{v},\vec{w}$ be such that $|\vec{u}|=1,\ |\vec{v}|=2,\ |\vec{w}|=3$. If the projection of $\vec{v}$ along $\vec{u}$ is equal to that of $\vec{w}$ along $\vec{u}$ and $\vec{v},\vec{w}$ are perpendicular to each other then $|\vec{u}-\vec{v}+\vec{w}|$ equals :

(A) 14 (B) $\sqrt{7}$ (C) $\sqrt{14}$ (D) 2

Open solution

Q182. AIEEE 2004

Let $\vec{a},\vec{b}$ and $\vec{c}$ be non-zero vectors such that $\left(\vec{a}\times\vec{b}\right)\times\vec{c}=\frac{1}{3}|\vec{b}||\vec{c}|\vec{a}$. If θ is the acute angle between the vectors $\vec{b}$ and $\vec{c}$, then $\sin\theta$ equals :

(A) $\frac{2\sqrt{2}}{3}$ (B) $\frac{\sqrt{2}}{3}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$

Open solution

Q183. AIEEE 2003

If $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$ then $\vec{a}+\vec{b}+\vec{c}=$

(A) abc (B) -1 (C) 0 (D) 2

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Q184. AIEEE 2003

Let $\vec{u} = \hat{i} + \hat{j}$, $\vec{v} = \hat{i} - \hat{j}$ and $\vec{w} = \hat{i} + 2\hat{j} + 3\hat{k}$. If $\hat{n}$ is a unit vector such that $\vec{u}.\hat{n} = 0$ and $\vec{v}.\hat{n} = 0$, then $\left|\vec{w}.\hat{n}\right|$ is equal to :

Open solution

(A) 3 (B) 0 (C) 1 (D) 2

Q185. AIEEE 2003

The vectors $\overrightarrow{AB} = 3\hat{i} + 4\hat{k}$ & $\overrightarrow{AC} = 5\hat{i} - 2\hat{j} + 4\hat{k}$ are the sides of triangle ABC. The length of the median through A is :

Open solution

(A) $\sqrt{288}$ (B) $\sqrt{18}$ (C) $\sqrt{72}$ (D) $\sqrt{33}$

Q186. AIEEE 2003

$\vec{a}, \vec{b}, \vec{c}$ are 3 vectors, such that

$\vec{a} + \vec{b} + \vec{c} = 0$, $\left|\vec{a}\right| = 1$ $\left|\vec{b}\right| = 2$, $\left|\vec{c}\right| = 3$,,

Open solution

then $\vec{a}.\vec{b} + \vec{b}.\vec{c} + \vec{c}.\vec{a}$ is equal to :

(A) 1 (B) 0 (C) -7 (D) 7

Q187. AIEEE 2003

A tetrahedron has vertices at $O(0, 0, 0)$, $A(1, 2, 1)$, $B(2, 1, 3)$ and $C(-1, 1, 2)$. Then the angle between the faces OAB and ABC will be :

Open solution

(A) 90° (B) $\cos^{-1}\left(\frac{19}{35}\right)$

(C) $\cos^{-1}\left(\frac{17}{31}\right)$ (D) 30°

Q188. AIEEE 2003

If $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$ and vectors $\left(1, a, a^2\right)$,

Open solution

$\left(1, b, b^2\right)$ and $\left(1, c, c^2\right)$ are non-coplanar, then the product abc equals :

(A) 0 (B) 2 (C) -1 (D) 1

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Q189. AIEEE 2003

Consider points A, B, C and D with position vectors $7\hat{i}-4\hat{j}+7\hat{k}, \hat{i}-6\hat{j}+10\hat{k}, -\hat{i}-3\hat{j}+4\hat{k}$ and $5\hat{i}-\hat{j}+5\hat{k}$ respectively. Then $ABCD$ is a :

(A) parallelogram but not a rhombus
(B) square
(C) rhombus
(D) None

Open solution

Q190. AIEEE 2002

If $\left|\vec{a}\right|=4, \left|\vec{b}\right|=2$ and the angle between $\vec{a}$ and $\vec{b}$ is $\pi/6$ then $\left(\vec{a}\times\vec{b}\right)^2$ is equal to :

(A) 48
(B) 16
(C) $\vec{a}$
(D) none of these

Open solution

Q191. AIEEE 2002

If the vectors $\vec{\mathbf{a}}, \vec{\mathbf{b}}$ and $\vec{\mathbf{c}}$ form the sides BC, CA and AB respectively of a triangle ABC, then :

(A) $\vec{\mathbf{a}}\cdot\vec{\mathbf{b}}=\vec{\mathbf{b}}\cdot\vec{\mathbf{c}}=\vec{\mathbf{c}}\cdot\vec{\mathbf{b}}=0$
(B) $\vec{\mathbf{a}}\times\vec{\mathbf{b}}=\vec{\mathbf{b}}\times\vec{\mathbf{c}}=\vec{\mathbf{c}}\times\vec{\mathbf{a}}$
(C) $\vec{\mathbf{a}}\cdot\vec{\mathbf{b}}=\vec{\mathbf{b}}\cdot\vec{\mathbf{c}}=\vec{\mathbf{c}}\cdot\vec{\mathbf{a}}=0$
(D) $\vec{\mathbf{a}}\times\vec{\mathbf{b}}+\vec{\mathbf{b}}\times\vec{\mathbf{c}}+\vec{\mathbf{c}}\times\vec{\mathbf{a}}=\vec{\mathbf{0}}$

Open solution

Q192. AIEEE 2002

If $\left|\vec{a}\right|=5, \left|\vec{b}\right|=4, \left|\vec{c}\right|=3$ thus what will be the value of $\left|\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}\right|$, given that $\vec{a}+\vec{b}+\vec{c}=0$:

(A) 25
(B) 50
(C) -25
(D) -50

Open solution

Q193. AIEEE 2002

If the vectors $\vec{c}, \vec{a}=x\hat{i}+y\hat{j}+z\hat{k}$ and $\hat{b}=\hat{j}$ are such that $\vec{a}, \vec{c}$ and $\vec{b}$ form a right handed system then $\vec{c}$ is :

(A) $z\hat{i}-x\hat{k}$
(B) $\vec{0}$
(C) $y\hat{j}$
(D) $-z\hat{i}+x\hat{k}$

Open solution

Q194. AIEEE 2002

$\vec{a}=3\hat{i}-5\hat{j}$ and $\vec{b}=6\hat{i}+3\hat{j}$ are two vectors and $\vec{c}$ is a vector such that $\vec{c}=\vec{a}\times\vec{b}$ then $\left|\vec{a}\right| : \left|\vec{b}\right| : \left|\vec{c}\right|$ =

(A) $\sqrt{34}:\sqrt{45}:\sqrt{39}$
(B) $\sqrt{34}:\sqrt{45}:39$
(C) $34:39:45$
(D) $39:35:34$

Open solution

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Numerical 48 questions

Q195. JEE Main 2026 (Online) 6th April Morning Shift

If $\vec{a}=\hat{i}+\hat{j}+\hat{k}, \vec{b}=\hat{j}-\hat{k}$ and $\vec{c}$ be three vectors such that $\vec{a}\times\vec{c}=\vec{b}$ and $\vec{a}\cdot\vec{c}=3$, then $\vec{c}\cdot(\vec{a}-2\vec{b})$ is equal to ____.

Open solution

Q196. JEE Main 2026 (Online) 28th January Morning Shift

Let PQR be a triangle such that $\overrightarrow{PQ}=-2\hat{i}-\hat{j}+2\hat{k}$ and $\overrightarrow{PR}=a\hat{i}+b\hat{j}-4\hat{k}, a,b\in Z$. Let S be the point on QR , which is equidistant from the lines PQ and PR . If $|\overrightarrow{PR}|=9$ and $\overrightarrow{PS}=\hat{i}-7\hat{j}+2\hat{k}$, then the value of $3a-4b$ is ____

Open solution

Q197. JEE Main 2026 (Online) 22nd January Evening Shift

Let a vector $\overrightarrow{a}=\sqrt{2}\hat{i}-\hat{j}+\lambda\hat{k}, \lambda>0$, make an obtuse angle with the vector $\overrightarrow{b}=-\lambda^2\hat{i}+4\sqrt{2}\hat{j}+4\sqrt{2}\hat{k}$ and an angle $\theta, \frac{\pi}{6}<\theta<\frac{\pi}{2}$, with the positive z-axis. If the set of all possible values of λ is $(\alpha,\beta)-\{\gamma\}$, then $\alpha+\beta+\gamma$ is equal to ____ .

Open solution

Q198. JEE Main 2025 (Online) 4th April Evening Shift

Let the three sides of a triangle ABC be given by the vectors $2\hat{i}-\hat{j}+\hat{k}, \hat{i}-3\hat{j}-5\hat{k}$ and $3\hat{i}-4\hat{j}-4\hat{k}$. Let G be the centroid of the triangle ABC. Then $6\left(|\overrightarrow{AG}|^2+|\overrightarrow{BG}|^2+|\overrightarrow{CG}|^2\right)$ is equal to _____.

Open solution

Q199. JEE Main 2025 (Online) 3rd April Evening Shift

Let $\vec{a}=\hat{i}+2\hat{j}+\hat{k}, \vec{b}=3\hat{i}-3\hat{j}+3\hat{k}, \vec{c}=2\hat{i}-\hat{j}+2\hat{k}$ and $\vec{d}$ be a vector such that $\vec{b}\times\vec{d}=\vec{c}\times\vec{d}$ and $\vec{a}\cdot\vec{d}=4$. Then $|(\vec{a}\times\vec{d})|^2$ is equal to_____.

Open solution

Q200. JEE Main 2025 (Online) 3rd April Morning Shift

Let $\vec{a}=\hat{i}+\hat{j}+\hat{k}, \vec{b}=3\hat{i}+2\hat{j}-\hat{k}, \vec{c}=\lambda\hat{j}+\mu\hat{k}$ and $\hat{d}$ be a unit vector such that $\vec{a}\times\hat{d}=\vec{b}\times\hat{d}$ and $\vec{c}\cdot\hat{d}=1$. If $\vec{c}$ is perpendicular to $\vec{a}$, then $|3\lambda\hat{d}+\mu\vec{c}|^2$ is equal to_____

Open solution

Q201. JEE Main 2025 (Online) 28th January Morning Shift

Let $\vec{a}=\hat{i}+\hat{j}+\hat{k}, \overrightarrow{b}=2\hat{i}+2\hat{j}+\hat{k}$ and $\overrightarrow{d}=\vec{a}\times\overrightarrow{b}$. If $\overrightarrow{c}$ is a vector such that $\vec{a}\cdot\overrightarrow{c}=|\overrightarrow{c}|, |\overrightarrow{c}-2\vec{a}|^2=8$ and the angle between $\overrightarrow{d}$ and $\overrightarrow{c}$ is $\frac{\pi}{4}$, then $|10-3\overrightarrow{b}\cdot\overrightarrow{c}|+|\overrightarrow{d}\times\overrightarrow{c}|^2$ is equal to _____.

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Q202. JEE Main 2025 (Online) 22nd January Morning Shift

Let $\vec{c}$ be the projection vector of $\vec{b} = \lambda\hat{i} + 4\hat{k}, \lambda > 0$, on the vector $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$. If $|\vec{a} + \vec{c}| = 7$, then the area of the parallelogram formed by the vectors $\vec{b}$ and $\vec{c}$ is _____.

Open solution

Q203. JEE Main 2024 (Online) 8th April Morning Shift

Let $\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}, \vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$ and $\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = (\vec{b} + \vec{c}) \times \vec{a}$ and $\vec{r} \cdot (\vec{b} - \vec{c}) = 0$, then $\frac{|593\vec{r} + 67\vec{a}|^2}{(593)^2}$ is equal to _____.

Open solution

Q204. JEE Main 2024 (Online) 6th April Morning Shift

Let $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}, \vec{b} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ and a vector $\vec{c}$ be such that $\vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times \vec{c} = \hat{i} + 8\hat{j} + 13\hat{k}$. If $\vec{a} \cdot \vec{c} = 13$, then $(24 - \vec{b} \cdot \vec{c})$ is equal to _____.

Open solution

Q205. JEE Main 2024 (Online) 5th April Morning Shift

Let $\vec{a} = \hat{i} - 3\hat{j} + 7\hat{k}, \vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $(\vec{a} + 2\vec{b}) \times \vec{c} = 3(\vec{c} \times \vec{a})$. If $\vec{a} \cdot \vec{c} = 130$, then $\vec{b} \cdot \vec{c}$ is equal to _____.

Open solution

Q206. JEE Main 2024 (Online) 4th April Morning Shift

Let ABC be a triangle of area $15\sqrt{2}$ and the vectors $\overrightarrow{AB} = \hat{i} + 2\hat{j} - 7\hat{k}, \overrightarrow{BC} = a\hat{i} + b\hat{j} + c\hat{k}$ and $\overrightarrow{AC} = 6\hat{i} + d\hat{j} - 2\hat{k}, d > 0$. Then the square of the length of the largest side of the triangle ABC is _____.

Open solution

Q207. JEE Main 2024 (Online) 1st February Evening Shift

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = -\hat{i} - 8\hat{j} + 2\hat{k}$ and $\vec{c} = 4\hat{i} + c_2\hat{j} + c_3\hat{k}$ be three vectors such that $\vec{b} \times \vec{a} = \vec{c} \times \vec{a}$. If the angle between the vector $\vec{c}$ and the vector $3\hat{i} + 4\hat{j} + \hat{k}$ is θ, then the greatest integer less than or equal to $\tan^2\theta$ is _____.

Open solution

Q208. JEE Main 2024 (Online) 31st January Evening Shift

Let $\vec{a} = 3\hat{i} + 2\hat{j} + \hat{k}, \vec{b} = 2\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{c}$ be a vector such that $(\vec{a} + \vec{b}) \times \vec{c} = 2(\vec{a} \times \vec{b}) + 24\hat{j} - 6\hat{k}$ and $(\vec{a} - \vec{b} + \hat{i}) \cdot \vec{c} = -3$. Then $|\vec{c}|^2$ is equal to _____.

Open solution

Q209. JEE Main 2024 (Online) 31st January Morning Shift

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}| = 1, |\vec{b}| = 4$, and $\vec{a} \cdot \vec{b} = 2$. If $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$ and the angle between $\vec{b}$ and $\vec{c}$ is α, then $192\sin^2\alpha$ is equal to _____.

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Q210. JEE Main 2024 (Online) 27th January Morning Shift

The least positive integral value of α, for which the angle between the vectors $\alpha\hat{i} - 2\hat{j} + 2\hat{k}$ and $\alpha\hat{i} + 2\alpha\hat{j} - 2\hat{k}$ is acute, is _____.

Open solution

Q211. JEE Main 2023 (Online) 13th April Morning Shift

Let $\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} - 3\hat{j} + 3\hat{k}$. If $\vec{b}$ is a vector such that $\vec{a} = \vec{b} \times \vec{c}$ and $|\vec{b}|^2 = 50$, then $|72 - |\vec{b} + \vec{c}|^2|$ is equal to _____.

Open solution

Q212. JEE Main 2023 (Online) 11th April Evening Shift

Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} + \hat{j} - \hat{k}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = 11, \vec{b} \cdot (\vec{a} \times \vec{c}) = 27$ and $\vec{b} \cdot \vec{c} = -\sqrt{3}|\vec{b}|$, then $|\vec{a} \times \vec{c}|^2$ is equal to _____.

Open solution

Q213. JEE Main 2023 (Online) 8th April Morning Shift

Let $\vec{a} = 6\hat{i} + 9\hat{j} + 12\hat{k}, \vec{b} = \alpha\hat{i} + 11\hat{j} - 2\hat{k}$ and $\vec{c}$ be vectors such that $\vec{a} \times \vec{c} = \vec{a} \times \vec{b}$. If

$\vec{a} \cdot \vec{c} = -12, \vec{c} \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 5$, then $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k})$ is equal to _____.

Open solution

Q214. JEE Main 2023 (Online) 1st February Morning Shift

$A(2, 6, 2), B(-4, 0, \lambda), C(2, 3, -1)$ and $D(4, 5, 0), |\lambda| \leq 5$ are the vertices of a quadrilateral $ABCD$. If its area is 18 square units, then $5 - 6\lambda$ is equal to _____.

Open solution

Q215. JEE Main 2023 (Online) 31st January Evening Shift

Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that

$|\vec{a}| = \sqrt{31}, 4|\vec{b}| = |\vec{c}| = 2$ and $2(\vec{a} \times \vec{b}) = 3(\vec{c} \times \vec{a})$.

If the angle between $\vec{b}$ and $\vec{c}$ is $\frac{2\pi}{3}$, then $\left(\frac{\vec{a} \times \vec{c}}{\vec{a} \cdot \vec{b}}\right)^2$ is equal to _____.

Open solution

Q216. JEE Main 2023 (Online) 31st January Morning Shift

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}| = \sqrt{14}, |\vec{b}| = \sqrt{6}$ and $|\vec{a} \times \vec{b}| = \sqrt{48}$. Then $(\vec{a} \cdot \vec{b})^2$ is equal to _____.

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Q217. JEE Main 2023 (Online) 24th January Evening Shift

Let $\vec{a}=\hat{i}+2\hat{j}+\lambda\hat{k}, \vec{b}=3\hat{i}-5\hat{j}-\lambda\hat{k}, \vec{a}.\vec{c}=7, 2\vec{b}.\vec{c}+43=0, \vec{a}\times\vec{c}=\vec{b}\times\vec{c}$. Then $\left|\vec{a}.\vec{b}\right|$ is equal to :

Open solution

Q218. JEE Main 2022 (Online) 30th July Evening Shift

Let $OABC$ be a parallelogram, O be the origin, $A(2,4,-5)$ and $C(b,2,3)$. If $P(a,a,a)$ and $Q(9-a^2,3,a-1), a\in\mathbb{N}$, are two points such that the projection of $\overrightarrow{OP}$ on $\overrightarrow{OB}$ is 2 and $\overrightarrow{OQ}$ makes acute angles with all three co-ordinate axes, then $\left|\overrightarrow{OB}\right|^2+\left|\overrightarrow{AC}\right|^2$ is equal to _____.

Open solution

Q219. JEE Main 2022 (Online) 29th July Evening Shift

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}+\vec{b}|^2=|\vec{a}|^2+2|\vec{b}|^2, \vec{a}\cdot\vec{b}=3$ and $|\vec{a}\times\vec{b}|^2=75$. Then $|\vec{a}|^2$ is equal to _____.

Open solution

Q220. JEE Main 2022 (Online) 27th July Evening Shift

Let $\vec{a},\vec{b},\vec{c}$ be three non-coplanar vectors such that $\vec{a}\times\vec{b}=4\vec{c}, \vec{b}\times\vec{c}=9\vec{a}$ and $\vec{c}\times\vec{a}=\alpha\vec{b}, \alpha>0$. If $\left|\vec{a}\right|+\left|\vec{b}\right|+\left|\vec{c}\right|=\frac{1}{36}$, then α is equal to _____.

Open solution

Q221. JEE Main 2022 (Online) 29th June Evening Shift

Let $\vec{a}=\hat{i}-2\hat{j}+3\hat{k}$, $\vec{b}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{c}$ be a vector such that $\vec{a}+\left(\vec{b}\times\vec{c}\right)=\vec{0}$ and $\vec{b}.\vec{c}=5$. Then the value of $3\left(\vec{c}.\vec{a}\right)$ is equal to _____.

Open solution

Q222. JEE Main 2022 (Online) 28th June Morning Shift

If $\vec{a}=2\hat{i}+\hat{j}+3\hat{k}, \vec{b}=3\hat{i}+3\hat{j}+\hat{k}$ and $\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$ are coplanar vectors and $\vec{a}.\vec{c}=5, \vec{b}\perp\vec{c}$, then $122(c_1+c_2+c_3)$ is equal to _____.

Open solution

Q223. JEE Main 2022 (Online) 25th June Evening Shift

Let $\vec{b}=\hat{i}+\hat{j}+\lambda\hat{k}$, $\lambda\in$ R. If $\vec{a}$ is a vector such that $\vec{a}\times\vec{b}=13\hat{i}-\hat{j}-4\hat{k}$ and $\vec{a}.\vec{b}+21=0$, then $\left(\vec{b}-\vec{a}\right).\left(\hat{k}-\hat{j}\right)+\left(\vec{b}+\vec{a}\right).\left(\hat{i}-\hat{k}\right)$ is equal to _____.

Open solution

Q224. JEE Main 2022 (Online) 25th June Morning Shift

Let θ be the angle between the vectors $\vec{a}$ and $\vec{b}$, where $|\vec{a}|=4$, $|\vec{b}|=3$ and $\theta\in\left(\frac{\pi}{4},\frac{\pi}{3}\right)$. Then $\left|\left(\vec{a}-\vec{b}\right)\times\left(\vec{a}+\vec{b}\right)\right|^2+4\left(\vec{a}.\vec{b}\right)^2$ is equal to _____.

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Q225. JEE Main 2021 (Online) 1st September Evening Shift

Let $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$. Let a vector $\vec{v}$ be in the plane containing $\vec{a}$ and $\vec{b}$. If $\vec{v}$ is perpendicular to the vector $3\hat{i} + 2\hat{j} - \hat{k}$ and its projection on $\vec{a}$ is 19 units, then $\left|2\vec{v}\right|^2$ is equal to _____.

Open solution

Q226. JEE Main 2021 (Online) 27th August Morning Shift

Let $\vec{a} = \hat{i} + 5\hat{j} + \alpha\hat{k}, \vec{b} = \hat{i} + 3\hat{j} + \beta\hat{k}$ and $\vec{c} = -\hat{i} + 2\hat{j} - 3\hat{k}$ be three vectors such that, $\left|\vec{b} \times \vec{c}\right| = 5\sqrt{3}$ and $\vec{a}$ is perpendicular to $\vec{b}$. Then the greatest amongst the values of $\left|\vec{a}\right|^2$ is _____.

Open solution

Q227. JEE Main 2021 (Online) 26th August Evening Shift

If the projection of the vector $\hat{i} + 2\hat{j} + \hat{k}$ on the sum of the two vectors $2\hat{i} + 4\hat{j} - 5\hat{k}$ and $-\lambda\hat{i} + 2\hat{j} + 3\hat{k}$ is 1, then λ is equal to _____.

Open solution

Q228. JEE Main 2021 (Online) 27th July Morning Shift

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b}$ and $\vec{c} = \hat{j} - \hat{k}$ be three vectors such that $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{a} \cdot \vec{b} = 1$. If the length of projection vector of the vector $\vec{b}$ on the vector $\vec{a} \times \vec{c}$ is l, then the value of $3l^2$ is equal to _____.

Open solution

Q229. JEE Main 2021 (Online) 25th July Evening Shift

If $\left(\vec{a} + 3\vec{b}\right)$ is perpendicular to $\left(7\vec{a} - 5\vec{b}\right)$ and $\left(\vec{a} - 4\vec{b}\right)$ is perpendicular to $\left(7\vec{a} - 2\vec{b}\right)$, then the angle between $\vec{a}$ and $\vec{b}$ (in degrees) is _____.

Open solution

Q230. JEE Main 2021 (Online) 25th July Morning Shift

Let $\vec{p} = 2\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{q} = \hat{i} + 2\hat{j} + \hat{k}$ be two vectors. If a vector $\vec{r} = (\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k})$ is perpendicular to each of the vectors $(\vec{p} + \vec{q})$ and $(\vec{p} - \vec{q})$, and $\left|\vec{r}\right| = \sqrt{3}$, then $|\alpha| + |\beta| + |\gamma|$ is equal to _____.

Open solution

Q231. JEE Main 2021 (Online) 20th July Evening Shift

For p > 0, a vector $\vec{v_2} = 2\hat{i} + (p+1)\hat{j}$ is obtained by rotating the vector $\vec{v_1} = \sqrt{3}p\hat{i} + \hat{j}$ by an angle θ about origin in counter clockwise direction. If $\tan\theta = \frac{\left(\alpha\sqrt{3}-2\right)}{\left(4\sqrt{3}+3\right)}$, then the value of α is equal to _____.

Open solution

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Q232. JEE Main 2021 (Online) 20th July Morning Shift

Let $\vec{a}, \vec{b}, \vec{c}$ be three mutually perpendicular vectors of the same magnitude and equally inclined at an angle θ, with the vector $\vec{a}+\vec{b}+\vec{c}$. Then $36\cos^2 2\theta$ is equal to _____.

Open solution

Q233. JEE Main 2021 (Online) 17th March Evening Shift

Let $\vec{x}$ be a vector in the plane containing vectors $\vec{a}=2\hat{i}-\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}-\hat{k}$. If the vector $\vec{x}$ is perpendicular to $\left(3\hat{i}+2\hat{j}-\hat{k}\right)$ and its projection on $\vec{a}$ is $\frac{17\sqrt{6}}{2}$, then the value of $|\vec{x}|^2$ is equal to _____.

Open solution

Q234. JEE Main 2021 (Online) 25th February Evening Shift

Let $\vec{a}=\hat{i}+\alpha\hat{j}+3\hat{k}$ and $\vec{b}=3\hat{i}-\alpha\hat{j}+\hat{k}$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $8\sqrt{3}$ square units, then $\vec{a}.\vec{b}$ is equal to _____.

Open solution

Q235. JEE Main 2021 (Online) 25th February Morning Shift

Let $\vec{a}=\hat{i}+2\hat{j}-\hat{k}, \vec{b}=\hat{i}-\hat{j}$ and $\vec{c}=\hat{i}-\hat{j}-\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r}\times\vec{a}=\vec{c}\times\vec{a}$ and $\vec{r}.\vec{b}=0$, then $\vec{r}.\vec{a}$ is equal to _____.

Open solution

Q236. JEE Main 2020 (Online) 6th September Evening Slot

If $\vec{x}$ and $\vec{y}$ be two non-zero vectors such that $\left|\vec{x}+\vec{y}\right|=\left|\vec{x}\right|$ and $2\vec{x}+\lambda\vec{y}$ is perpendicular to $\vec{y}$, then the value of λ is _____ .

Open solution

Q237. JEE Main 2020 (Online) 6th September Morning Slot

If $\vec{a}$ and $\vec{b}$ are unit vectors, then the greatest value of

$\sqrt{3}\left|\vec{a}+\vec{b}\right|+\left|\vec{a}-\vec{b}\right|$ is_____.

Open solution

Q238. JEE Main 2020 (Online) 5th September Evening Slot

Let the vectors $\vec{a}, \vec{b}, \vec{c}$ be such that $\left|\vec{a}\right|=2, \left|\vec{b}\right|=4$ and $\left|\vec{c}\right|=4$. If the projection of $\vec{b}$ on $\vec{a}$ is equal to the projection of $\vec{c}$ on $\vec{a}$ and $\vec{b}$ is perpendicular to $\vec{c}$, then the value of $\left|\vec{a}+\vec{b}-\vec{c}\right|$ is _____.

Open solution

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Q239. JEE Main 2020 (Online) 4th September Evening Slot

If $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$, then the value of

$\left|\hat{i} \times \left(\vec{a} \times \hat{i}\right)\right|^2 + \left|\hat{j} \times \left(\vec{a} \times \hat{j}\right)\right|^2 + \left|\hat{k} \times \left(\vec{a} \times \hat{k}\right)\right|^2$ is equal to____

Open solution

Q240. JEE Main 2020 (Online) 2nd September Evening Slot

Let the position vectors of points 'A' and 'B' be $\hat{i} + \hat{j} + \hat{k}$ and $2\hat{i} + \hat{j} + 3\hat{k}$, respectively. A point 'P' divides the line segment AB internally in the ratio λ : 1 (λ > 0). If O is the origin and $\overrightarrow{OB}.\overrightarrow{OP} - 3\left|\overrightarrow{OA} \times \overrightarrow{OP}\right|^2 = 6$, then λ is equal to_____.

Open solution

Q241. JEE Main 2020 (Online) 2nd September Morning Slot

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three unit vectors such that $\left|\vec{a} - \vec{b}\right|^2 + \left|\vec{a} - \vec{c}\right|^2 = 8$.

Then $\left|\vec{a} + 2\vec{b}\right|^2 + \left|\vec{a} + 2\vec{c}\right|^2$ is equal to _____.

Open solution

Q242. JEE Main 2020 (Online) 9th January Evening Slot

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors such that $\left|\vec{a}\right| = \sqrt{3}$, $\left|\vec{b}\right| = 5, \vec{b}.\vec{c} = 10$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$. If $\vec{a}$ is perpendicular to the vector $\vec{b} \times \vec{c}$, then $\left|\vec{a} \times \left(\vec{b} \times \vec{c}\right)\right|$ is equal to _____.

Open solution

Out of syllabus 65 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q243. JEE Main 2023 (Online) 15th April Morning Shift · MCQ (Single Correct Answer)

Let S be the set of all (λ, μ) for which the vectors $\lambda\hat{i} - \hat{j} + \hat{k}, \hat{i} + 2\hat{j} + \mu\hat{k}$ and $3\hat{i} - 4\hat{j} + 5\hat{k}$, where $\lambda - \mu = 5$, are coplanar, then $\sum\limits_{(\lambda,\mu)\in S} 80\left(\lambda^2 + \mu^2\right)$ is equal to :

(A) 2370 (B) 2130 (C) 2210 (D) 2290

Open solution

Q244. JEE Main 2023 (Online) 12th April Morning Shift · MCQ (Single Correct Answer)

Let a, b, c be three distinct real numbers, none equal to one. If the vectors $a\hat{i} + \hat{j} + \hat{k}, \hat{i} + b\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} + c\hat{k}$ are coplanar, then $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ is equal to :

(A) -2 (B) 1 (C) -1 (D) 2

Open solution

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Q245. JEE Main 2023 (Online) 12th April Morning Shift · MCQ (Single Correct Answer)

Let $\lambda \in \mathbb{Z}, \vec{a} = \lambda\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$. Let $\vec{c}$ be a vector such that $(\vec{a} + \vec{b} + \vec{c}) \times \vec{c} = \vec{0}, \vec{a} \cdot \vec{c} = -17$ and $\vec{b} \cdot \vec{c} = -20$. Then $|\vec{c} \times (\lambda\hat{i} + \hat{j} + \hat{k})|^2$ is equal to :

(A) 53 (B) 62 (C) 49 (D) 46

Open solution

Q246. JEE Main 2023 (Online) 11th April Evening Shift · MCQ (Single Correct Answer)

If four distinct points with position vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are coplanar, then $[\vec{a}\ \vec{b}\ \vec{c}]$ is equal to :

(A) $[\vec{d}\ \ \vec{b}\ \ \vec{a}] + [\vec{a}\ \ \vec{c}\ \ \vec{d}] + [\vec{d}\ \ \vec{b}\ \ \vec{c}]$

(B) $[\vec{b}\ \ \vec{c}\ \ \vec{d}] + [\vec{d}\ \ \vec{a}\ \ \vec{c}] + [\vec{d}\ \ \vec{b}\ \ \vec{a}]$

(C) $[\vec{a}\ \ \vec{d}\ \ \vec{b}] + [\vec{d}\ \ \vec{c}\ \ \vec{a}] + [\vec{d}\ \ \vec{b}\ \ \vec{c}]$

(D) $[\vec{d}\ \ \vec{c}\ \ \vec{a}] + [\vec{b}\ \ \vec{d}\ \ \vec{a}] + [\vec{c}\ \ \vec{d}\ \ \vec{b}]$

Open solution

Q247. JEE Main 2023 (Online) 8th April Evening Shift · MCQ (Single Correct Answer)

Let the vectors $\vec{u}_1 = \hat{i} + \hat{j} + a\hat{k}, \vec{u}_2 = \hat{i} + b\hat{j} + \hat{k}$ and $\vec{u}_3 = c\hat{i} + \hat{j} + \hat{k}$ be coplanar. If the vectors $\vec{v}_1 = (a+b)\hat{i} + c\hat{j} + c\hat{k}, \vec{v}_2 = a\hat{i} + (b+c)\hat{j} + a\hat{k}$ and $\vec{v}_3 = b\hat{i} + b\hat{j} + (c+a)\hat{k}$ are also coplanar, then $6(\mathrm{a} + \mathrm{b} + \mathrm{c})$ is equal to :

(A) 12 (B) 6 (C) 0 (D) 4

Open solution

Q248. JEE Main 2023 (Online) 6th April Evening Shift · MCQ (Single Correct Answer)

Let the vectors $\vec{a}, \vec{b}, \vec{c}$ represent three coterminous edges of a parallelopiped of volume V. Then the volume of the parallelopiped, whose coterminous edges are represented by $\vec{a}, \vec{b} + \vec{c}$ and $\vec{a} + 2\vec{b} + 3\vec{c}$ is equal to :

(A) 3 V (B) 2 V (C) 6 V (D) V

Open solution

Q249. JEE Main 2023 (Online) 6th April Evening Shift · MCQ (Single Correct Answer)

The sum of all values of α, for which the points whose position vectors are $\hat{i} - 2\hat{j} + 3\hat{k}, 2\hat{i} - 3\hat{j} + 4\hat{k}, (\alpha+1)\hat{i} + 2\hat{k}$ and $9\hat{i} + (\alpha - 8)\hat{j} + 6\hat{k}$ are coplanar, is equal to :

(A) 6 (B) 4 (C) -2 (D) 2

Open solution

Q250. JEE Main 2023 (Online) 6th April Morning Shift · MCQ (Single Correct Answer)

Let the position vectors of the points A, B, C and D be $5\hat{i} + 5\hat{j} + 2\lambda\hat{k}, \hat{i} + 2\hat{j} + 3\hat{k}, -2\hat{i} + \lambda\hat{j} + 4\hat{k}$ and $-\hat{i} + 5\hat{j} + 6\hat{k}$. Let the set $S = \{\lambda \in \mathbb{R} :$ the points A, B, C and D are coplanar $\}$.

Then $\sum_{\lambda \in S} (\lambda + 2)^2$ is equal to :

(A) $\frac{37}{2}$ (B) 25 (C) 13 (D) 41

Open solution

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Q251. JEE Main 2023 (Online) 30th January Evening Shift · MCQ (Single Correct Answer)

Let $\lambda \in \mathbb{R}, \vec{a} = \lambda\hat{i} + 2\hat{j} - 3\hat{k}, \vec{b} = \hat{i} - \lambda\hat{j} + 2\hat{k}$.

If $((\vec{a} + \vec{b}) \times (\vec{a} \times \vec{b})) \times (\vec{a} - \vec{b}) = 8\hat{i} - 40\hat{j} - 24\hat{k}$,

then $|\lambda(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})|^2$ is equal to :

(A) 136 (B) 140 (C) 144 (D) 132

Open solution

Q252. JEE Main 2023 (Online) 30th January Morning Shift · MCQ (Single Correct Answer)

If $\vec{a}, \vec{b}, \vec{c}$ are three non-zero vectors and $\hat{n}$ is a unit vector perpendicular to $\vec{c}$ such that $\vec{a} = \alpha\vec{b} - \hat{n}, (\alpha \neq 0)$ and $\vec{b} \cdot \vec{c} = 12$, then $\left|\vec{c} \times (\vec{a} \times \vec{b})\right|$ is equal to :

(A) 15 (B) 9 (C) 6 (D) 12

Open solution

Q253. JEE Main 2023 (Online) 25th January Evening Shift · MCQ (Single Correct Answer)

Let $\vec{a} = -\hat{i} - \hat{j} + \hat{k}, \vec{a} \cdot \vec{b} = 1$ and $\vec{a} \times \vec{b} = \hat{i} - \hat{j}$. Then $\vec{a} - 6\vec{b}$ is equal to :

(A) $3\left(\hat{i} + \hat{j} + \hat{k}\right)$ (B) $3\left(\hat{i} - \hat{j} - \hat{k}\right)$

(C) $3\left(\hat{i} + \hat{j} - \hat{k}\right)$ (D) $3\left(\hat{i} - \hat{j} + \hat{k}\right)$

Open solution

Q254. JEE Main 2023 (Online) 25th January Evening Shift · MCQ (Single Correct Answer)

If the four points, whose position vectors are $3\hat{i} - 4\hat{j} + 2\hat{k}, \hat{i} + 2\hat{j} - \hat{k}, -2\hat{i} - \hat{j} + 3\hat{k}$ and $5\hat{i} - 2\alpha\hat{j} + 4\hat{k}$ are coplanar, then α is equal to :

(A) $\frac{73}{17}$ (B) $-\frac{73}{17}$ (C) $-\frac{107}{17}$ (D) $\frac{107}{17}$

Open solution

Q255. JEE Main 2023 (Online) 25th January Morning Shift · MCQ (Single Correct Answer)

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non zero vectors such that $\vec{b} \cdot \vec{c} = 0$ and $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b} - \vec{c}}{2}$. If $\vec{d}$ be a vector such that $\vec{b} \cdot \vec{d} = \vec{a} \cdot \vec{b}$, then $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d})$ is equal to

(A) $\frac{1}{2}$ (B) $-\frac{1}{4}$ (C) $\frac{1}{4}$ (D) $\frac{3}{4}$

Open solution

Q256. JEE Main 2023 (Online) 24th January Morning Shift · MCQ (Single Correct Answer)

Let $\vec{u} = \hat{i} - \hat{j} - 2\hat{k}, \vec{v} = 2\hat{i} + \hat{j} - \hat{k}, \vec{v} \cdot \vec{w} = 2$ and $\vec{v} \times \vec{w} = \vec{u} + \lambda\vec{v}$. Then $\vec{u} \cdot \vec{w}$ is equal to :

(A) $-\frac{2}{3}$ (B) $\frac{3}{2}$ (C) 2 (D) 1

Open solution

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Q257. JEE Main 2022 (Online) 29th July Morning Shift · MCQ (Single Correct Answer)

Let $\vec{a}=3\hat{i}+\hat{j}$ and $\vec{b}=\hat{i}+2\hat{j}+\hat{k}$. Let $\vec{c}$ be a vector satisfying $\vec{a}\times(\vec{b}\times\vec{c})=\vec{b}+\lambda\vec{c}$. If $\vec{b}$ and $\vec{c}$ are non-parallel, then the value of λ is :

(A) -5 (B) 5 (C) 1 (D) -1

Open solution

Q258. JEE Main 2022 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

Let $\vec{a}=2\hat{i}-\hat{j}+5\hat{k}$ and $\vec{b}=\alpha\hat{i}+\beta\hat{j}+2\hat{k}$. If $((\vec{a}\times\vec{b})\times\hat{i})\cdot\hat{k}=\frac{23}{2}$, then $|\vec{b}\times 2\hat{j}|$ is equal to :

(A) 4 (B) 5 (C) $\sqrt{21}$ (D) $\sqrt{17}$

Open solution

Q259. JEE Main 2022 (Online) 26th July Evening Shift · MCQ (Single Correct Answer)

A vector $\vec{a}$ is parallel to the line of intersection of the plane determined by the vectors $\hat{i},\hat{i}+\hat{j}$ and the plane determined by the vectors $\hat{i}-\hat{j},\hat{i}+\hat{k}$. The obtuse angle between $\vec{a}$ and the vector $\vec{b}=\hat{i}-2\hat{j}+2\hat{k}$ is :

(A) $\frac{3\pi}{4}$ (B) $\frac{2\pi}{3}$ (C) $\frac{4\pi}{5}$ (D) $\frac{5\pi}{6}$

Open solution

Q260. JEE Main 2022 (Online) 30th June Morning Shift · MCQ (Single Correct Answer)

Let a vector $\vec{c}$ be coplanar with the vectors $\vec{a}=-\hat{i}+\hat{j}+\hat{k}$ and $\vec{b}=2\hat{i}+\hat{j}-\hat{k}$. If the vector $\vec{c}$ also satisfies the conditions $\vec{c}.\left[\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}\times\vec{b}\right)\right]=-42$ and $\left(\vec{c}\times\left(\vec{a}-\vec{b}\right)\right).\hat{k}=3$, then the value of $|\vec{c}|^2$ is equal to :

(A) 24 (B) 29 (C) 35 (D) 42

Open solution

Q261. JEE Main 2022 (Online) 26th June Morning Shift · MCQ (Single Correct Answer)

If $\vec{a}.\vec{b}=1,\vec{b}.\vec{c}=2$ and $\vec{c}.\vec{a}=3$, then the value of $\left[\vec{a}\times\left(\vec{b}\times\vec{c}\right),\vec{b}\times\left(\vec{c}\times\vec{a}\right),\vec{c}\times\left(\vec{b}\times\vec{a}\right)\right]$ is :

(A) 0

(B) $-6\vec{a}.\left(\vec{b}\times\vec{c}\right)$

(C) $-12\vec{c}.\left(\vec{a}\times\vec{b}\right)$

(D) $-12\vec{b}.\left(\vec{c}\times\vec{a}\right)$

Open solution

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Q262. JEE Main 2021 (Online) 31st August Evening Shift · MCQ (Single Correct Answer)

Open solution

Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors mutually perpendicular to each other and have same magnitude. If a vector $\vec{r}$ satisfies

$\vec{a} \times \{(\vec{r} - \vec{b}) \times \vec{a}\} + \vec{b} \times \{(\vec{r} - \vec{c}) \times \vec{b}\} + \vec{c} \times \{(\vec{r} - \vec{a}) \times \vec{c}\} = \vec{0}$, then $\vec{r}$ is equal to :

(A) $\frac{1}{3}(\vec{a} + \vec{b} + \vec{c})$

(B) $\frac{1}{3}(2\vec{a} + \vec{b} - \vec{c})$

(C) $\frac{1}{2}(\vec{a} + \vec{b} + \vec{c})$

(D) $\frac{1}{2}(\vec{a} + \vec{b} + 2\vec{c})$

Q263. JEE Main 2021 (Online) 26th August Morning Shift · MCQ (Single Correct Answer)

Open solution

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{j} - \hat{k}$. If $\vec{c}$ is a vector such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$, then $\vec{a} \cdot (\vec{b} \times \vec{c})$ is equal to :

(A) -2

(B) -6

(C) 6

(D) 2

Q264. JEE Main 2021 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

Open solution

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors such that $\vec{a} = \vec{b} \times (\vec{b} \times \vec{c})$. If magnitudes of the vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are $\sqrt{2}$, 1 and 2 respectively and the angle between $\vec{b}$ and $\vec{c}$ is θ $\left(0 < \theta < \frac{\pi}{2}\right)$, then the value of 1 + tanθ is equal to :

(A) $\sqrt{3} + 1$

(B) 2

(C) 1

(D) $\frac{\sqrt{3}+1}{\sqrt{3}}$

Q265. JEE Main 2021 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

Open solution

Let the vectors

$(2 + a + b)\hat{i} + (a + 2b + c)\hat{j} - (b + c)\hat{k}, (1 + b)\hat{i} + 2b\hat{j} - b\hat{k}$ and $(2 + b)\hat{i} + 2b\hat{j} + (1 - b)\hat{k}, a, b, c \in R$

be co-planar. Then which of the following is true?

(A) 2b = a + c

(B) 3c = a + b

(C) a = b + 2c

(D) 2a = b + c

Q266. JEE Main 2021 (Online) 22th July Evening Shift · MCQ (Single Correct Answer)

Open solution

Let a vector $\vec{a}$ be coplanar with vectors $\vec{b} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} + \hat{k}$. If $\vec{a}$ is perpendicular to $\vec{d} = 3\hat{i} + 2\hat{j} + 6\hat{k}$, and $\left|\vec{a}\right| = \sqrt{10}$. Then a possible value of $[\vec{a}\ \vec{b}\ \vec{c}] + [\vec{a}\ \vec{b}\ \vec{d}] + [\vec{a}\ \vec{c}\ \vec{d}]$ is equal to :

(A) -42

(B) -40

(C) -29

(D) -38

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Q267. JEE Main 2021 (Online) 22th July Evening Shift · MCQ (Single Correct Answer)

Let three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ be such that $\vec{a} \times \vec{b} = \vec{c}, \vec{b} \times \vec{c} = \vec{a}$ and $|\vec{a}| = 2$. Then which one of the following is not true?

(A) $\vec{a} \times \left((\vec{b} + \vec{c}) \times (\vec{b} - \vec{c}) \right) = \vec{0}$

(B) Projection of $\vec{a}$ on $(\vec{b} \times \vec{c})$ is 2

(C) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] + \left[\vec{c}\ \vec{a}\ \vec{b}\right] = 8$

(D) $\left|3\vec{a} + \vec{b} - 2\vec{c}\right|^2 = 51$

Open solution

Q268. JEE Main 2021 (Online) 20th July Morning Shift · MCQ (Single Correct Answer)

Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $\vec{a}.\vec{c} = |\vec{c}|, |\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $(\vec{a} \times \vec{b})$ and $\vec{c}$ is $\frac{\pi}{6}$, then the value of $\left| \left(\vec{a} \times \vec{b} \right) \times \vec{c} \right|$ is :

(A) $\frac{2}{3}$ (B) 4 (C) 3 (D) $\frac{3}{2}$

Open solution

Q269. JEE Main 2021 (Online) 17th March Morning Shift · MCQ (Single Correct Answer)

Let $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{b} = 7\hat{i} + \hat{j} - 6\hat{k}$.

If $\vec{r} \times \vec{a} = \vec{r} \times \vec{b}, \vec{r}.(\hat{i} + 2\hat{j} + \hat{k}) = -3$, then $\vec{r}.(2\hat{i} - 3\hat{j} + \hat{k})$ is equal to :

(A) 10 (B) 8 (C) 13 (D) 12

Open solution

Q270. JEE Main 2021 (Online) 26th February Morning Shift · MCQ (Single Correct Answer)

If $\vec{a}$ and $\vec{b}$ are perpendicular, then $\vec{a} \times \left(\vec{a} \times \left(\vec{a} \times \left(\vec{a} \times \vec{b} \right) \right) \right)$ is equal to :

(A) $\frac{1}{2}|\vec{a}|^4\vec{b}$

(B) $\vec{0}$

(C) $\vec{a} \times \vec{b}$

(D) $|\vec{a}|^4\vec{b}$

Open solution

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Q271. JEE Main 2020 (Online) 5th September Morning Slot · MCQ (Single Correct Answer)

Open solution

If the volume of a parallelopiped, whose coterminus edges are given by the vectors $\vec{a} = \hat{i} + \hat{j} + n\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} - n\hat{k}$ and $\vec{c} = \hat{i} + n\hat{j} + 3\hat{k}$ $(n \geq 0)$, is 158 cu. units, then :

(A) n = 7

(B) $\vec{b}.\vec{c} = 10$

(C) $\vec{a}.\vec{c} = 17$

(D) n = 9

Q272. JEE Main 2020 (Online) 4th September Morning Slot · MCQ (Single Correct Answer)

Open solution

Let x_0 be the point of Local maxima of $f(x) = \vec{a}.\left(\vec{b} \times \vec{c}\right)$, where $\vec{a} = x\hat{i} - 2\hat{j} + 3\hat{k}, \vec{b} = -2\hat{i} + x\hat{j} - \hat{k}, \vec{c} = 7\hat{i} - 2\hat{j} + x\hat{k}$. Then the value of $\vec{a}.\vec{b} + \vec{b}.\vec{c} + \vec{c}.\vec{a}$ at $x = x_0$ is :

(A) 14 (B) -30 (C) -4 (D) -22

Q273. JEE Main 2020 (Online) 8th January Morning Slot · MCQ (Single Correct Answer)

Open solution

Let the volume of a parallelopiped whose coterminous edges are given by

$\vec{u} = \hat{i} + \hat{j} + \lambda\hat{k}, \vec{v} = \hat{i} + \hat{j} + 3\hat{k}$ and

$\vec{w} = 2\hat{i} + \hat{j} + \hat{k}$ be 1 cu. unit. If θ be the angle between the edges $\vec{u}$ and $\vec{w}$, then $\cos\theta$ can be :

(A) $\frac{7}{6\sqrt{3}}$ (B) $\frac{7}{6\sqrt{6}}$ (C) $\frac{5}{7}$ (D) $\frac{5}{3\sqrt{3}}$

Q274. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

Open solution

Let $\alpha \in$ R and the three vectors

$\vec{a} = \alpha\hat{i} + \hat{j} + 3\hat{k}, \vec{b} = 2\hat{i} + \hat{j} - \alpha\hat{k}$

and $\vec{c} = \alpha\hat{i} - 2\hat{j} + 3\hat{k}$.

Then the set S = $\{\alpha : \vec{a}, \vec{b}$ and $\vec{c}$ are coplanar$\}$:

(A) contains exactly two numbers only one of which is positive

(B) is singleton

(C) contains exactly two positive numbers

(D) is empty

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Q275. JEE Main 2019 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

If the volume of parallelopiped formed by the vectors $\hat{i}+\lambda\hat{j}+\hat{k}$, $\hat{j}+\lambda\hat{k}$ and $\lambda\hat{i}+\hat{k}$ is minimum, then λ is equal to :

(A) $-\frac{1}{\sqrt{3}}$ (B) $\sqrt{3}$ (C) $-\sqrt{3}$ (D) $\frac{1}{\sqrt{3}}$

Open solution

Q276. JEE Main 2019 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The magnitude of the projection of the vector $2\hat{i}+3\hat{j}+\hat{k}$ on the vector perpendicular to the plane containing the vectors $\hat{i}+\hat{j}+\hat{k}$ and $\hat{i}+2\hat{j}+3\hat{k}$, is :

(A) $\frac{\sqrt{3}}{2}$ (B) $\sqrt{6}$ (C) $\sqrt{\frac{3}{2}}$ (D) $3\sqrt{6}$

Open solution

Q277. JEE Main 2019 (Online) 12th January Evening Slot · MCQ (Single Correct Answer)

Let $\vec{a},\vec{b}$ and $\vec{c}$ be three unit vectors, out of which vectors $\vec{b}$ and $\vec{c}$ are non-parallel. If α and β are the angles which vector $\vec{a}$ makes with vectors $\vec{b}$ and $\vec{c}$ respectively and $\vec{a}\times(\vec{b}\times\vec{c})=\frac{1}{2}\vec{b}$, then $|\alpha-\beta|$ is equal to :

(A) 90° (B) 30° (C) 45° (D) 60°

Open solution

Q278. JEE Main 2019 (Online) 12th January Morning Slot · MCQ (Single Correct Answer)

The sum of the distinct real values of μ, for which the vectors, $\mu\hat{i}+\hat{j}+\hat{k}$, $\hat{i}+\mu\hat{j}+\hat{k}$, $\hat{i}+\hat{j}+\mu\hat{k}$ are co-planar, is :

(A) 2 (B) -1 (C) 0 (D) 1

Open solution

Q279. JEE Main 2019 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

Let $\vec{a}=\hat{i}-\hat{j}$, $\vec{b}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{c}$

be a vector such that $\vec{a}\times\vec{c}+\vec{b}=\vec{0}$

and $\vec{a}\cdot\vec{c}=4$, then $|\vec{c}|^2$ is equal to :

(A) 8 (B) $\frac{19}{2}$ (C) 9 (D) $\frac{17}{2}$

Open solution

Q280. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

Let $\vec{a},\vec{b}$ and $\vec{c}$ be three unit vectors such that $\vec{a}\times\left(\vec{b}\times\vec{c}\right)=\frac{\sqrt{3}}{2}\left(\vec{b}+\vec{c}\right)$. If $\vec{b}$ is not parallel to $\vec{c}$, then the angle between $\vec{a}$ and $\vec{b}$ is:

(A) $\frac{2\pi}{3}$ (B) $\frac{5\pi}{6}$ (C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{2}$

Open solution

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Q281. JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-zero vectors such that no two of them are collinear and $\left(\vec{a} \times \vec{b}\right) \times \vec{c} = \frac{1}{3}\left|\vec{b}\right|\left|\vec{c}\right|\vec{a}$. If θ is the angle between vectors $\vec{b}$ and $\vec{c}$, then a value of $\sin\theta$ is :

(A) $\frac{2}{3}$ (B) $\frac{-2\sqrt{3}}{3}$ (C) $\frac{2\sqrt{2}}{3}$ (D) $\frac{-\sqrt{2}}{3}$

Open solution

Q282. JEE Main 2014 (Offline) · MCQ (Single Correct Answer)

If $\left[\vec{a} \times \vec{b}\ \ \vec{b} \times \vec{c}\ \ \vec{c} \times \vec{a}\right] = \lambda\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]^2$ then λ is equal to :

(A) 0 (B) 1 (C) 2 (D) 3

Open solution

Q283. JEE Main 2013 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

The vector $\left(\hat{i} \times \vec{a}.\vec{b}\right)\hat{i} + \left(\hat{j} \times \vec{a}.\vec{b}\right)\hat{j} + \left(\hat{k} \times \vec{a}.\vec{b}\right)\hat{k}$ is equal to :

(A) $\vec{b} \times \vec{a}$

(B) $\vec{a}$

(C) $\vec{a} \times \vec{b}$

(D) $\vec{b}$

Open solution

Q284. AIEEE 2012 (Online) 26th May · MCQ (Single Correct Answer)

Statement 1: The vectors $\vec{a}, \vec{b}$ and $\vec{c}$ lie in the same plane if and only if $\vec{a} \cdot (\vec{b} \times \vec{c}) = 0$
Statement 2: The vectors $\vec{u}$ and $\vec{v}$ are perpendicular if and only if $\vec{u} \cdot \vec{v} = 0$ where $\vec{u} \times \vec{v}$ is a vector perpendicular to the plane of $\vec{u}$ and $\vec{v}$

(A) Statement 1 is false, Statement 2 is true.

(B) Statement 1 is true, Statement 2 is true, Statement 2 is correct explanation for Statement 1.

(C) Statement 1 is true, Statement 2 is false.

(D) Statement 1 is true, Statement 2 is true, Statement 2 is not a correct explanation for Statement 1.

Open solution

Q285. AIEEE 2012 (Online) 19th May · MCQ (Single Correct Answer)

If $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}, \vec{b} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{c} = r\hat{i} + \hat{j} + (2r - 1)\hat{k}$ are three vectors such that $\vec{c}$ is parallel to the plane of $\vec{a}$ and $\vec{b}$, then r is equal to

(A) 1 (B) -1 (C) 0 (D) 2

Open solution

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Q286. AIEEE 2012 (Online) 12th May · MCQ (Single Correct Answer)

If $\vec{u} = \hat{j} + 4\hat{k}, \vec{v} = \hat{i} + 3\hat{k}$ and $\vec{w} = \cos\theta\hat{i} + \sin\theta\hat{j}$ are vectors in 3-dimensional space, then the maximum possible value of $|\vec{u} \times \vec{v} \cdot \vec{w}|$ is

(A) $\sqrt{3}$ (B) 5 (C) $\sqrt{14}$ (D) 7

Open solution

Q287. AIEEE 2012 (Online) 12th May · MCQ (Single Correct Answer)

Statement 1: If the points $(1, 2, 2), (2, 1, 2)$ and $(2, 2, z)$ and $(1, 1, 1)$ are coplanar, then $z = 2$.
Statement 2: If the 4 points P, Q, R and S are coplanar, then the volume of the tetrahedron $PQRS$ is 0.

(A) Statement 1 is false, Statement 2 is true.

(B) Statement 1 is true, Statement 2 is false.

(C) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation of Statement 1.

(D) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation of Statement 1.

Open solution

Q288. AIEEE 2012 (Online) 12th May · MCQ (Single Correct Answer)

A unit vector which is perpendicular to the vector $2\hat{i} - \hat{j} + 2\hat{k}$ and is coplanar with the vectors $\hat{i} + \hat{j} - \hat{k}$ and $2\hat{i} + 2\hat{j} - \hat{k}$ is

(A) $\frac{2\hat{j}+\hat{k}}{\sqrt{5}}$ (B) $\frac{3\hat{i}+2\hat{j}-2\hat{k}}{\sqrt{17}}$

(C) $\frac{3\hat{i}+2\hat{j}+2\hat{k}}{\sqrt{17}}$ (D) $\frac{2\hat{i}+2\hat{j}-\hat{k}}{3}$

Open solution

Q289. AIEEE 2012 (Online) 7th May · MCQ (Single Correct Answer)

If $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}, \vec{b} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{c} = \lambda\hat{i} + \hat{j} + (2\lambda - 1)\hat{k}$ are coplanar vectors, then λ is equal to

(A) 0 (B) -1 (C) 2 (D) 1

Open solution

Q290. AIEEE 2011 · MCQ (Single Correct Answer)

If $\overrightarrow{a} = \frac{1}{\sqrt{10}}\left(3\hat{i} + \hat{k}\right)$ and $\overrightarrow{b} = \frac{1}{7}\left(2\hat{i} + 3\hat{j} - 6\hat{k}\right)$, then the value of $\left(2\overrightarrow{a} - \overrightarrow{b}\right).\left[\left(\overrightarrow{a} \times \overrightarrow{b}\right) \times \left(\overrightarrow{a} + 2\overrightarrow{b}\right)\right]$ is :

(A) -3 (B) 5 (C) 3 (D) -5

Open solution

Q291. AIEEE 2010 · MCQ (Single Correct Answer)

Let $\overrightarrow{a} = \hat{j} - \hat{k}$ and $\overrightarrow{c} = \hat{i} - \hat{j} - \hat{k}$. Then the vector $\overrightarrow{b}$ satisfying $\overrightarrow{a} \times \overrightarrow{b} + \overrightarrow{c} = \overrightarrow{0}$ and $\overrightarrow{a}.\overrightarrow{b} = 3$:

(A) $2\hat{i} - \hat{j} + 2\hat{k}$ (B) $\hat{i} - \hat{j} - 2\hat{k}$

(C) $\hat{i} + \hat{j} - 2\hat{k}$ (D) $-\hat{i} + \hat{j} - 2\hat{k}$

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Q292. AIEEE 2009 · MCQ (Single Correct Answer)

If $\vec{u}, \vec{v}, \vec{w}$ are non-coplanar vectors and p, q are real numbers, then the equality $\left[3\vec{u}\ p\vec{v}\ p\vec{w}\right] - \left[p\vec{v}\ \vec{w}\ q\vec{u}\right] - \left[2\vec{w}\ q\vec{v}\ q\vec{u}\right] = 0$ holds for :

Open solution

(A) exactly two values of (p, q)

(B) more than two but not all values of (p, q)

(C) all values of (p, q)

(D) exactly one value of (p, q)

Q293. AIEEE 2008 · MCQ (Single Correct Answer)

The vector $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$ lies in the plane of the vectors $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{j} + \hat{k}$ and bisects the angle between $\vec{b}$ and $\vec{c}$. Then which one of the following gives possible values of α and β ?

Open solution

(A) $\alpha = 2,\ \beta = 2$

(B) $\alpha = 1,\ \beta = 2$

(C) $\alpha = 2,\ \beta = 1$

(D) $\alpha = 1,\ \beta = 1$

Q294. AIEEE 2007 · MCQ (Single Correct Answer)

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{i} - \hat{j} + 2\hat{k}$ and $\vec{c} = x\hat{i} + (x-2)\hat{j} - \hat{k}$. If the vectors $\vec{c}$ lies in the plane of $\vec{a}$ and $\vec{b}$, then x equals :

Open solution

(A) -4

(B) -2

(C) 0

(D) 1.

Q295. AIEEE 2006 · MCQ (Single Correct Answer)

If $\left(\vec{a} \times \vec{b}\right) \times \vec{c} = \vec{a} \times \left(\vec{b} \times \vec{c}\right)$ where $\vec{a}, \vec{b}$ and $\vec{c}$ are any three vectors such that $\vec{a}.\vec{b} \neq 0,\ \vec{b}.\vec{c} \neq 0$ then $\vec{a}$ and $\vec{c}$ are :

Open solution

(A) inclined at an angle of $\frac{\pi}{3}$ between them

(B) inclined at an angle of $\frac{\pi}{6}$ between them

(C) perpendicular

(D) parallel

Q296. AIEEE 2005 · MCQ (Single Correct Answer)

If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar vectors and λ is a real number then

$\left[\lambda\left(\vec{a}+\vec{b}\right)\quad \lambda^2\vec{b}\quad \lambda\vec{c}\right] = \left[\vec{a}\quad \vec{b}+\vec{c}\quad \vec{b}\right]$ for :

Open solution

(A) exactly one value of λ

(B) no value of λ

(C) exactly three values of λ

(D) exactly two values of λ

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Q297. AIEEE 2005 · MCQ (Single Correct Answer)

Let $\vec{a} = \hat{i} - \hat{k}$, $\vec{b} = x\hat{i} + \hat{j} + (1-x)\hat{k}$ and $\vec{c} = y\hat{i} + x\hat{j} + (1+x-y)\hat{k}$. Then $\left[\vec{a}, \vec{b}, \vec{c}\right]$ depends on :

(A) only y (B) only x (C) both x and y (D) neither x nor y

Open solution

Q298. AIEEE 2004 · MCQ (Single Correct Answer)

If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors and λ is a real number, then the vectors $\vec{a} + 2\vec{b} + 3\vec{c}$, $\lambda\vec{b} + 4\vec{c}$ and $(2\lambda - 1)\vec{c}$ are non coplanar for :

(A) no value of λ (B) all except one value of λ

(C) all except two values of λ (D) all values of λ

Open solution

Q299. AIEEE 2003 · MCQ (Single Correct Answer)

If $\vec{u}, \vec{v}$ and $\vec{w}$ are three non-coplanar vectors, then $\left(\vec{u} + \vec{v} - \vec{w}\right) \cdot \left(\vec{u} - \vec{v}\right) \times \left(\vec{v} - \vec{w}\right)$ equals :

(A) $3\vec{u} \cdot \vec{v} \times \vec{w}$ (B) 0

(C) $\vec{u} \cdot \vec{v} \times \vec{w}$ (D) $\vec{u} \cdot \vec{w} \times \vec{v}$

Open solution

Q300. AIEEE 2002 · MCQ (Single Correct Answer)

If $\vec{a}$, $\vec{b}$, $\vec{c}$ are vectors such that $\left[\vec{a}\,\vec{b}\,\vec{c}\right] = 4$ then $\left[\vec{a} \times \vec{b}\ \vec{b} \times \vec{c}\ \vec{c} \times \vec{a}\right] =$

(A) 16 (B) 64 (C) 4 (D) 8

Open solution

Q301. JEE Main 2023 (Online) 1st February Morning Shift · Numerical

Let $\vec{v} = \alpha\hat{i} + 2\hat{j} - 3\hat{k}, \vec{w} = 2\alpha\hat{i} + \hat{j} - \hat{k}$ and $\vec{u}$ be a vector such that $|\vec{u}| = \alpha > 0$. If the minimum value of the scalar triple product $\left[\vec{u}\ \vec{v}\ \vec{w}\right]$ is $-\alpha\sqrt{3401}$, and $|\vec{u} \cdot \hat{i}|^2 = \frac{m}{n}$ where m and n are coprime natural numbers, then $m + n$ is equal to _____.

Open solution

Q302. JEE Main 2023 (Online) 29th January Morning Shift · Numerical

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-zero non-coplanar vectors. Let the position vectors of four points A, B, C and D be $\vec{a} - \vec{b} + \vec{c}, \lambda\vec{a} - 3\vec{b} + 4\vec{c}, -\vec{a} + 2\vec{b} - 3\vec{c}$ and $2\vec{a} - 4\vec{b} + 6\vec{c}$ respectively. If $\overrightarrow{AB}, \overrightarrow{AC}$ and $\overrightarrow{AD}$ are coplanar, then λ is equal to _____.

Open solution

Q303. JEE Main 2021 (Online) 27th July Evening Shift · Numerical

Let $\vec{a} = \hat{i} - \alpha\hat{j} + \beta\hat{k}$, $\vec{b} = 3\hat{i} + \beta\hat{j} - \alpha\hat{k}$ and $\vec{c} = -\alpha\hat{i} - 2\hat{j} + \hat{k}$, where α and β are integers. If $\vec{a} \cdot \vec{b} = -1$ and $\vec{b} \cdot \vec{c} = 10$, then $\left(\vec{a} \times \vec{b}\right) \cdot \vec{c}$ is equal to _____.

Open solution

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Q304. JEE Main 2021 (Online) 17th March Morning Shift · **Numerical**

If $\vec{a} = \alpha\hat{i} + \beta\hat{j} + 3\hat{k}$,

$\vec{b} = -\beta\hat{i} - \alpha\hat{j} - \hat{k}$ and

$\vec{c} = \hat{i} - 2\hat{j} - \hat{k}$

such that $\vec{a} . \vec{b} = 1$ and $\vec{b} . \vec{c} = -3$, then $\frac{1}{3}\left(\left(\vec{a} \times \vec{b}\right) . \vec{c}\right)$ is equal to _____.

Open solution

Q305. JEE Main 2021 (Online) 16th March Evening Shift · **Numerical**

Let $\vec{c}$ be a vector perpendicular to the vectors, $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. If $\vec{c} . \left(\hat{i} + \hat{j} + 3\hat{k}\right) = 8$ then the value of $\vec{c} . \left(\vec{a} \times \vec{b}\right)$ is equal to _____.

Open solution

Q306. JEE Main 2021 (Online) 24th February Morning Shift · **Numerical**

Let three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ be such that $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$, $\vec{a} . \vec{c} = 7$ and $\vec{b}$ is perpendicular to $\vec{c}$, where $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{k}$, then the value of $2\left|\vec{a} + \vec{b} + \vec{c}\right|^2$ is _____.

Open solution

Q307. JEE Main 2020 (Online) 9th January Morning Slot · **Numerical**

If the vectors, $\vec{p} = (a+1)\hat{i} + a\hat{j} + a\hat{k}$,

$\vec{q} = a\hat{i} + (a+1)\hat{j} + a\hat{k}$ and

$\vec{r} = a\hat{i} + a\hat{j} + (a+1)\hat{k}\,(a \in R)$

are coplanar and $3\left(\vec{p} . \vec{q}\right)^2 - \lambda\left|\vec{r} \times \vec{q}\right|^2 = 0$, then the value of λ is _____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **C**	3. **B**	4. **A**	5. **A**
6. **B**	7. **B**	8. **C**	9. **B**	10. **B**
11. *Scan QR*	12. **C**	13. **A**	14. **C**	15. **D**
16. **B**	17. **D**	18. **A**	19. **A**	20. **C**
21. **D**	22. **A**	23. **A**	24. **B**	25. **D**
26. **A**	27. **D**	28. **C**	29. **C**	30. **C**
31. **D**	32. **B**	33. **D**	34. **A**	35. **C**
36. **A**	37. **D**	38. **D**	39. **D**	40. **C**
41. **A**	42. **C**	43. **B**	44. **B**	45. **B**
46. **C**	47. **C**	48. **A**	49. **C**	50. **A**
51. **D**	52. **A**	53. **C**	54. **C**	55. **A**
56. **A**	57. **B**	58. **B**	59. **C**	60. **D**
61. **D**	62. **B**	63. **C**	64. **D**	65. **C**
66. **D**	67. **A**	68. **C**	69. **C**	70. **D**
71. **A**	72. **B**	73. **D**	74. **D**	75. **C**
76. **B**	77. **B**	78. **D**	79. **B**	80. **C**
81. **B**	82. **A**	83. **D**	84. **C**	85. **C**
86. **A**	87. **A**	88. **C**	89. **D**	90. **A**
91. **D**	92. **B**	93. **D**	94. **C**	95. **A**
96. **B**	97. **C**	98. **B**	99. **D**	100. **D**
101. **A**	102. **C**	103. **A**	104. **A**	105. **D**
106. **C**	107. **D**	108. **A**	109. **D**	110. **B**
111. **C**	112. **C**	113. **C**	114. **D**	115. **B**
116. **D**	117. **A**	118. **C**	119. **D**	120. **C**
121. **D**	122. **B**	123. **D**	124. **B**	125. **A**
126. **D**	127. **A**	128. **C**	129. **B**	130. **A**
131. **A**	132. **C**	133. **B**	134. **C**	135. **B**
136. **B**	137. **D**	138. **C**	139. **B**	140. **C**
141. **A**	142. **C**	143. **A**	144. **B**	145. **B**
146. **A**	147. **D**	148. **C**	149. **B**	150. **A**

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151. **C**	152. **B**	153. **D**	154. **C**	155. **D**
156. **B**	157. **B**	158. **B**	159. **C**	160. **A**
161. **D**	162. **C**	163. **B**	164. **C**	165. **D**
166. **D**	167. **D**	168. **B**	169. **C**	170. **A**
171. **A**	172. **D**	173. **B**	174. **A**	175. **A**
176. **C**	177. **B**	178. **D**	179. **D**	180. **A**
181. **C**	182. **A**	183. **C**	184. **A**	185. **D**
186. **C**	187. **B**	188. **C**	189. **D**	190. **B**
191. **B**	192. **A**	193. **A**	194. **B**	

Numerical

195. **3**	196. **37**	197. **5**	198. **164**	199. **128**
200. **5**	201. **6**	202. **16**	203. **569**	204. **46**
205. **30**	206. **54**	207. **38**	208. **38**	209. **48**
210. **5**	211. **66**	212. **285**	213. **11**	214. **11**
215. **3**	216. **36**	217. **8**	218. **118**	219. **14**
220. **36**	221. **BONUS**	222. **150**	223. **14**	224. **576**
225. **1494**	226. **90**	227. **5**	228. **2**	229. **60**
230. **3**	231. **6**	232. **4**	233. **486**	234. **2**
235. **12**	236. **1**	237. **4**	238. **6**	239. **18**
240. **0.8**	241. **2**	242. **30**		

Out of syllabus

243. **D**	244. **B**	245. **D**	246. **D**	247. **A**
248. **D**	249. **D**	250. **D**	251. **B**	252. **D**
253. **A**	254. **A**	255. **C**	256. **D**	257. **A**
258. **B**	259. **A**	260. **C**	261. **A**	262. **C**
263. **A**	264. **B**	265. **A**	266. **A**	267. **D**
268. **D**	269. **D**	270. **D**	271. **B**	272. **D**
273. **A**	274. **D**	275. **D**	276. **C**	277. **B**
278. **B**	279. **B**	280. **B**	281. **C**	282. **B**
283. **C**	284. **B**	285. **C**	286. **B**	287. **A**
288. **D**	289. **A**	290. **D**	291. **D**	292. **D**
293. **D**	294. **B**	295. **D**	296. **B**	297. **D**

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298. **C**	299. **C**	300. **A**	301. **3501**	302. **2**
303. **9**	304. **2**	305. **28**	306. **75**	307. **1**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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